Forget, but Not Lost: Strong Forgetting for Expressive Ontologies

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Abstract

Ontologies serve as the schema layer for knowledge graphs and knowledge-intensive systems, providing structured, machine interpretable semantics essential for data integration, information access, and intelligent knowledge management. However, as these ontologies grow in scale and expressivity, their simplification, modularization, and reuse have become formidable challenges. Strong forgetting, a model-preserving technique for ontology refinement, offers a theoretically ideal solution. Yet, due to its inherent complexity, existing methods struggle to handle the sophisticated constructors prevalent in modern, expressive ontologies. Notably, no practical strong forgetting method exists for logics that incorporate both qualified number restrictions ($\mathcal{Q}$) and inverse roles ($\mathcal{I}$). This paper addresses this critical gap by presenting the first practical method for strong role forgetting in ontologies specified in the description logic $\mathcal{ALCQI}(\nabla)$. Our approach tackles the elimination of role names, a task more challenging than, yet subsuming, concept forgetting. The core of our method is a novel, sound, and terminating inference calculus that precisely handles the complex interplay between qualified number restrictions and inverse roles during elimination, incorporating the universal role (∇) when necessary to avoid information loss. We have conducted extensive empirical evaluations on large-scale, real-world ontologies from the Oxford-ISG and BioPortal benchmarks. The results demonstrate that despite the inherent undecidability of the problem, our approach achieves high success rates and superb efficiency in practice, paving the way for scalable knowledge management, curation, and reuse of complex ontological knowledge bases. Our implementation, experimental data, and an extended version with full proofs and additional analyses are available at https://anonymous.4open.science/r/alcqi/.

CCS Concepts

• **Computing methodologies → Ontology engineering**.

Keywords

Ontology, Description Logic, Strong Forgetting, Knowledge Reuse

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1 Introduction

Large-scale knowledge graphs and ontologies have become foundational infrastructure for modern knowledge management, powering applications from intelligent information retrieval and data integration to knowledge-driven AI systems [2]. Ontologies formalized in description logics (DLs) provide the structured, machine-interpretable semantics that underpin the Web Ontology Language OWL 2 DL [10], the W3C standard for publishing and exchanging ontological knowledge. As these ontologies grow in scale and expressivity, the ability to manage, refine, and reuse them becomes a pressing challenge for the knowledge management community.

A central operation in ontology management is *forgetting* [19], which eliminates designated symbols from an ontology while retaining all information independent of them. Forgetting has deep roots in logic, tracing back to Boole's variable elimination [3] and Ackermann's second-order quantifier elimination [1]. In the context of DL ontologies, it serves as the formal foundation for a range of knowledge management tasks: extracting self-contained ontology modules for knowledge reuse [9], debugging faulty ontologies [25, 26], aligning independently developed ontologies [17, 31], and computing logical differences between ontology versions [11, 20, 39].

Two variants of forgetting have been studied in the literature. *Weak forgetting* [18, 21, 35, 38] adopts a deductive view: the result preserves all entailments expressible over the remaining signature. *Strong forgetting* [19, 37] provides a strictly stronger, model-theoretic guarantee: it preserves the entire model structure modulo the forgotten symbols. This distinction has practical consequences. Generating a faithful view of an ontology for knowledge sharing, for instance, requires not merely that the view agrees with the original on entailments, but that it admits exactly the right set of models [9]. Strong forgetting also enables richer explanations in ontology-based query answering [6, 15], since it captures model-level rather than merely deductive relationships. Theoretically, strong forgetting coincides with second-order quantifier elimination [1, 8], for which general-purpose methods such as SCAN [7], DLS [28], and SQEMA [5] have been developed at the level of classical logic.

Despite its clear advantages, strong forgetting is remarkably difficult to realize in practice. Determining whether a finite strong forgetting result even exists is undecidable for DLs as simple as $\mathcal{EL}$ and $\mathcal{ALC}$ [12]. When a result does exist, its size can be triple exponential relative to the input [22, 24]. These worst-case barriers have channeled most practical work toward weak forgetting or toward logics of limited expressivity [29, 33]. The two most mature forgetting systems, LETHE and FAME, support role forgetting for $\mathcal{ALCI}$ and $\mathcal{ALCOIH}$ respectively [41, 43, 44], but neither handles qualified number restrictions. Recent work [32] addresses strong forgetting for $\mathcal{ALCQ}$, yet it does not support inverse roles. No existing method handles both $\mathcal{Q}$ and $\mathcal{I}$ simultaneously.

This limitation constitutes a critical gap. Qualified number restrictions ($\mathcal{Q}$) and inverse roles ($\mathcal{I}$) are among the most widely used constructors in real-world ontologies. Qualified number restrictions express cardinality constraints such as "every clinical trial must involve at least two investigators," while inverse roles enable bidirectional navigation of relationships, as when hasPart and isPartOf are declared as inverses of each other. The DL $\mathcal{ALCQI}$, which combines both constructors, is a core fragment of OWL 2 DL and appears prevalently in biomedical, enterprise, and scientific knowledge bases. Yet no practical method for strong forgetting exists for this logic. Existing tools either support $\mathcal{Q}$ without $\mathcal{I}$, or $\mathcal{I}$ without $\mathcal{Q}$, reverting to weak forgetting when both are present [41, 43].

The technical obstacles behind this gap are substantial, and the step from $\mathcal{ALCQ}$ to $\mathcal{ALCQI}$ represents a fundamental complexity boundary rather than an incremental extension. In $\mathcal{ALCQ}$ (without inverse roles), constraints propagate only in a single direction along the role structure: from a domain element to its role successors. This unidirectional flow enables a divide-and-conquer strategy in which processed sub-structures can be discarded once their contributions have been resolved. Introducing inverse roles fundamentally disrupts this locality. A constraint such as $\forall r^-.C$ propagates information *backwards* from successors to predecessors, creating bidirectional dependencies that prevent any local decomposition. At the level of model theory, the combination of $\mathcal{Q}$ and $\mathcal{I}$ destroys the tree model property enjoyed by $\mathcal{ALCQ}$, elevating the worst-case complexity of key reasoning tasks from ExpTime to NExpTime-hard [30].

For forgetting, these differences manifest in the inference mechanism. In $\mathcal{ALCQ}$, eliminating a role name r involves resolving $\geq r.$-literals against $\leq r.$-literals, which constitutes a single type of arithmetic interaction. In $\mathcal{ALCQI}$, the calculus must simultaneously handle four distinct types of interactions: forward against forward ($\geq r.$ vs. $\leq r.$), inverse against inverse ($\geq r^-.$ vs. $\leq r^-.$), and two cross-interactions between forward and inverse orientations ($\geq r.$ vs. $\geq r^-.$ and $\leq r^-.$ vs. $\leq r.$). The cross-role interactions are particularly challenging: they involve arithmetic reasoning about cardinality constraints that propagate in opposite directions, a pattern that has no analogue in $\mathcal{ALCQ}$ and cannot be handled by any straightforward extension of existing methods. Our calculus resolves these interactions through purely syntactic inference rules, avoiding the need for semantic entailment checks that would be intractable in $\mathcal{ALCQI}$. A further complication arises from the fact that the forgetting result may fall outside $\mathcal{ALCQI}$ itself; in such cases the universal role ∇ must be introduced to avoid information loss.

This paper presents the first practical method for strong role forgetting in $\mathcal{ALCQI}(\nabla)$. We focus on role forgetting because it subsumes concept forgetting through a well-known reduction [43]: any concept name A can be eliminated by introducing a fresh role r, replacing A with $\exists r.\top$, and then forgetting r. **Our contributions**:

- We design a sound and terminating inference calculus that eliminates role names from $\mathcal{ALCQI}(\nabla)$-ontologies. It is the first calculus to handle the full interplay between qualified number restrictions and inverse roles in a strong forgetting context.
- We implement a prototype and evaluate it extensively on 338 real-world ontologies from two benchmark repositories. Despite the theoretical undecidability of the problem, our method achieves success rates of up to 90% and runs 3.6 to 4.8 times faster than the state of the art, while consuming significantly less memory.
- We provide detailed experimental analyses covering output size, the impact of inverse roles, universal role usage frequency, and phase-level time breakdowns, yielding comprehensive insights into the practical behavior of forgetting on real-world ontologies.

2 Preliminaries

We begin by defining the syntax and semantics of the DL $\mathcal{ALCQI}$, which serves as the foundation for our work. Let $\mathsf{N_C}$ and $\mathsf{N_R}$ be countably infinite, disjoint sets of concept and role names, respectively. A *role* in $\mathcal{ALCQI}$ is either a role name $r \in \mathsf{N_R}$ or an *inverse role* r^- of a role name r, while $\mathcal{ALCQI}(\nabla)$ extends this to include the universal role ∇. *Concept descriptions* in $\mathcal{ALCQI}$ (or simply *concepts*) are constructed according to the following syntax:

$$\top \mid A \mid \neg C \mid C \sqcap D \mid C \sqcup D \mid \geq mR.C \mid \leq nR.C,$$

where $A \in \mathsf{N_C}$, C and D range over concepts, R ranges over roles, and $m \geq 1$ and $n \geq 0$ are natural numbers. We use standard abbreviations: $\bot = \neg\top$, $\exists R.C = \geq 1R.C$, $\forall R.C = \leq 0R.\neg C$, $\neg{\geq}mR.C = {\leq}nR.C$ and $\neg{\leq}nR.C = {\geq}mR.C$ with $n = m-1$. Concepts of the form $\geq mR.C$ and $\leq nR.C$ are called *qualified number restrictions*, which express cardinality constraints on role relationships.

An $\mathcal{ALCQI}$-ontology $\mathcal{O}$ is a finite set of *axioms* of the form $C \sqsubseteq D$, called *general concept inclusions* (or simply *GCIs*), where C and D are concepts. We use $C \equiv D$ as an abbreviation for the pair of GCIs $C \sqsubseteq D$ and $D \sqsubseteq C$.

The semantics of $\mathcal{ALCQI}$ is defined in terms of an *interpretation* $\mathcal{I} = \langle \Delta^{\mathcal{I}}, \cdot^{\mathcal{I}} \rangle$, where $\Delta^{\mathcal{I}}$ is a non-empty set called the *domain*, and $\cdot^{\mathcal{I}}$ is an *interpretation function* that maps every concept name $A \in \mathsf{N_C}$ to a set $A^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}}$, and every role name $r \in \mathsf{N_R}$ to a binary relation $r^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}}$. The interpretation function is inductively extended to complex concepts and inverse roles as follows:

$$\top^{\mathcal{I}} = \Delta^{\mathcal{I}} \qquad \nabla^{\mathcal{I}} = \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}} \qquad (\neg C)^{\mathcal{I}} = \Delta^{\mathcal{I}} \setminus C^{\mathcal{I}}$$
$$(C \sqcap D)^{\mathcal{I}} = C^{\mathcal{I}} \cap D^{\mathcal{I}} \qquad (C \sqcup D)^{\mathcal{I}} = C^{\mathcal{I}} \cup D^{\mathcal{I}}$$
$$(\geq mR.C)^{\mathcal{I}} = \{x \in \Delta^{\mathcal{I}} \mid \#\{(x,y) \in R^{\mathcal{I}} \mid y \in C^{\mathcal{I}}\} \geq m\}$$
$$(\leq nR.C)^{\mathcal{I}} = \{x \in \Delta^{\mathcal{I}} \mid \#\{(x,y) \in R^{\mathcal{I}} \mid y \in C^{\mathcal{I}}\} \leq n\}$$
$$(R^-)^{\mathcal{I}} = \{(y,x) \in \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}} \mid (x,y) \in R^{\mathcal{I}}\}$$

where $\#\{\cdot\}$ denotes the cardinality of a set.

A GCI $C \sqsubseteq D$ is *true* in an interpretation $\mathcal{I}$ iff $C^{\mathcal{I}} \subseteq D^{\mathcal{I}}$. $\mathcal{I}$ is a *model* of an ontology $\mathcal{O}$, written $\mathcal{I} \models \mathcal{O}$, iff every GCI in $\mathcal{O}$ is true in $\mathcal{I}$. A GCI $C \sqsubseteq D$ is a *logical consequence* of $\mathcal{O}$, written $\mathcal{O} \models C \sqsubseteq D$, iff $C \sqsubseteq D$ is true in every model of $\mathcal{O}$.

Our forgetting method operates on GCIs in clausal normal form. A *literal* is a concept in one of the following forms: A, $\neg A$, $\geq mR.C$, or $\leq nR.C$, where $A \in \mathsf{N_C}$, R is a role (possibly inverse), $m \geq 1$ and $n \geq 0$ are natural numbers, and C is a concept. A *clause* is a GCI of the form $\top \sqsubseteq L_1 \sqcup \ldots \sqcup L_n$, where each L_i (for $1 \leq i \leq n$) is a literal. Following standard practice, we omit the prefix '$\top \sqsubseteq$' and treat clauses as sets, so they contain no duplicates and their order is irrelevant. Thus, stating that $C^{\mathcal{I}} \cup (\geq mr.D)^{\mathcal{I}}$ is true means that $\top \sqsubseteq C \sqcup (\geq mr.D)$ is true in $\mathcal{I}$. A GCI can be converted into clausal form using standard transformations [2].

Let $\mathcal{S} \in \mathsf{N_C} \cup \mathsf{N_R}$ be a designated name. A clause containing $\mathcal{S}$ is called an $\mathcal{S}$-*clause*. Within an $\mathcal{S}$-clause, an occurrence of $\mathcal{S}$ is *positive* if it appears under an even number of (explicit and implicit) negations, and *negative* if under an odd number. Translation to first-order logic reveals all implicit negations. For example, in the clause $\geq 2r.A$, the role r occurs positively, as shown by its first-order representation $\forall x \exists y, z(r(x,y) \wedge r(x,z) \wedge A(y) \wedge A(z) \wedge y \neq z)$. In contrast, in the clause $\leq 1r^{-}.A$, the inverse role r^{-} occurs negatively, as evidenced by its first-order form $\forall x, y, z(\neg r(y,x) \vee \neg r(z,x) \vee \neg A(y) \vee \neg A(z) \vee y = z)$.

Given two interpretations $\mathcal{I}$ and $\mathcal{I}'$, we say that $\mathcal{I}$ and $\mathcal{I}'$ are $\mathcal{S}$*-equivalent* (denoted $\mathcal{I} \sim_{\mathcal{S}} \mathcal{I}'$) if they differ at most in the interpretation of $\mathcal{S}$. This generalizes to $\mathcal{F}$*-equivalence* (denoted $\mathcal{I} \sim_{\mathcal{F}} \mathcal{I}'$) for a set $\mathcal{F}$ of role names, meaning that $\mathcal{I}$ and $\mathcal{I}'$ may differ only in how they interpret names in $\mathcal{F}$. Formally, $\mathcal{I} \sim_{\mathcal{F}} \mathcal{I}'$ holds iff:

(i) $\mathcal{I}$ and $\mathcal{I}'$ have the same domain ($\Delta^{\mathcal{I}} = \Delta^{\mathcal{I}'}$) and interpret each concept name $A \in \mathsf{N_C}$ identically ($A^{\mathcal{I}} = A^{\mathcal{I}'}$);
(ii) for each role name $r \in \mathsf{N_R} \setminus \mathcal{F}$, we have $r^{\mathcal{I}} = r^{\mathcal{I}'}$.

The *signature* $\mathrm{sig}(\mathcal{O})$ of an ontology $\mathcal{O}$ is the set of all concept and role names occurring in $\mathcal{O}$, and $\mathrm{sig_R}(\mathcal{O}) = \mathrm{sig}(\mathcal{O}) \cap \mathsf{N_R}$ denotes its *role signature.*

Definition 1 (**Strong Forgetting for $\mathcal{ALCQI}(\nabla)$**). *Let $\mathcal{O}$ be an $\mathcal{ALCQI}(\nabla)$-ontology and $\mathcal{F} \subseteq sig_R(\mathcal{O})$ be a set of role names to be forgotten, called the* forgetting signature. *An ontology $\mathcal{V}$ is a* result of forgetting $\mathcal{F}$ from $\mathcal{O}$ *if:*

(i) *$sig(\mathcal{V}) \subseteq sig(\mathcal{O}) \setminus \mathcal{F}$, and*
(ii) *for any interpretation $\mathcal{I}'$, we have $\mathcal{I}' \models \mathcal{V}$ iff there exists an interpretation $\mathcal{I} \sim_{\mathcal{F}} \mathcal{I}'$ such that $\mathcal{I} \models \mathcal{O}$.*

Two important properties follow from this definition:

(i) Decomposability: Forgetting $\mathcal{F}$ from $\mathcal{O}$ can be achieved by eliminating individual names in $\mathcal{F}$ one at a time, in any order.
(ii) Uniqueness up to equivalence: Any two results $\mathcal{V}$ and $\mathcal{V}'$ of forgetting $\mathcal{F}$ from $\mathcal{O}$ are logically equivalent, though they may differ syntactically. This allows us to speak definitively of *the* result of forgetting, rather than indefinitely *a* result.

3 Ontology Normalization

Our forgetting method is designed to eliminate a single role name at a time. To facilitate this, we first transform the ontology into a specific normal form with respect to the target role name to be forgotten. Our method takes an $\mathcal{ALCQI}(\nabla)$-ontology as input and operates entirely within this logic throughout the computation.

The calculus operates on a dedicated normal form designed for $\mathcal{ALCQI}(\nabla)$-ontologies, which isolates occurrences of a role name r and its inverse r^{-} into a standardized structure.

Definition 2 (r-**Normal Form**). *Let $r \in N_R$ be a role name. A clause is in r-*normal form *(or r-*NF *for short) if it contains r or r^{-} and has one of the following forms:*

$$C \sqcup \geq mR.D \quad \text{or} \quad E \sqcup \leq nR.F$$

*where $R \in \{r, r^{-}\}$, and the concepts C, D, E, and F do not contain any occurrence of r or r^{-}. An $\mathcal{ALCQI}(\nabla)$-ontology $\mathcal{O}$ is in r-*NF *if every clause in $\mathcal{O}$ containing r or r^{-} is in r-NF.*

The normal form establishes a canonical structure where each relevant clause contains exactly one occurrence of the target role name (or its inverse), located at the top level of a qualified number restriction. This syntactic uniformity is essential for the subsequent inference steps.

Any unnormalized $\mathcal{ALCQI}(\nabla)$-ontology $\mathcal{O}$ can be transformed into r-NF by exhaustively applying the normalization rules NR1–NR3. This transformation process introduces fresh concept names, termed *definers* [16], which act as abbreviations for complex concepts, thereby untangling nested structures. For a role name r to be forgotten, the rules are applied to any clause not in r-NF because r or r^{-} appears in a nested position.

NR1 For a clause $C \sqcup \geq mS.D$ or $E \sqcup \leq nS.F$ not in r-NF, if the sub-concept C (or E) contains an occurrence of r or r^{-}, replace C (or E) with a fresh definer $Z \in \mathsf{N_C}$ and add the new clause $\neg Z \sqcup C$ (or $\neg Z \sqcup E$) to $\mathcal{O}$.

NR2 For a clause $C \sqcup \geq mS.D$ not in r-NF, if the sub-concept D contains an occurrence of r or r^{-}, replace D with a fresh definer $Z \in \mathsf{N_C}$ and add the new clause $\neg Z \sqcup D$ to $\mathcal{O}$.

NR3 For a clause $E \sqcup \leq nS.F$ not in r-NF, if the sub-concept F contains an occurrence of r or r^{-}, replace F with a fresh definer $Z \in \mathsf{N_C}$ and add the new clause $Z \sqcup \neg F$ to $\mathcal{O}$.

In practice, multiple normalization rules may be applicable to the same clause. Our method addresses this by sequentially applying NR1–NR3 in order. Importantly, the order of rule application does not affect the correctness of the final result, as the normalization process is confluent.

Example 1. *Consider the following $\mathcal{ALCQI}$-ontology $\mathcal{O}$ with a single complex clause, and let the forgetting signature be $\mathcal{F} = \{r\}$.*

$$\{1.\ \leq 1r.\geq 2r^{-}.A_1 \sqcup \geq 2s.\leq 1t.\leq 3r.A_2 \sqcup \leq 0t.A_3\}$$

This clause is not in r-NF because the role r and its inverse r^{-} appear in multiple, nested positions. Specifically, $\geq 2r^{-}.A_1$ is nested within a $\leq 1r$ restriction, and $\leq 3r.A_2$ is nested within $\leq 1t$ and $\geq 2s$ restrictions. Our method applies the normalization rules sequentially to isolate each of these occurrences.
Applying NR1 to Clause 1 to handle the right-hand part of the disjunction ($Z_1 \in N_C$ is a fresh definer):

$$\{2.\ \leq 1r.\geq 2r^{-}.A_1 \sqcup Z_1 \quad 3.\ \neg Z_1 \sqcup \geq 2s.\leq 1t.\leq 3r.A_2 \sqcup \leq 0t.A_3\}$$

Applying NR3 to Clause 2 to resolve the nested expression containing r^{-} ($Z_2 \in N_C$ is a fresh definer):

$$\{4.\ \leq 1r.Z_2 \sqcup Z_1 \quad 5.\ Z_2 \sqcup \neg(\geq 2r^{-}.A_1)$$
$$3.\ \neg Z_1 \sqcup \geq 2s.\leq 1t.\leq 3r.A_2 \sqcup \leq 0t.A_3\}$$

Applying NR2 to Clause 3 to isolate the sub-concept containing the next occurrence of r ($Z_3 \in N_C$ is a fresh definer):

$$\{4.\ \leq 1r.Z_2 \sqcup Z_1 \quad 5.\ Z_2 \sqcup \leq 1r^{-}.A_1$$
$$6.\ \neg Z_1 \sqcup \geq 2s.Z_3 \sqcup \leq 0t.A_3 \quad 7.\ \neg Z_3 \sqcup \leq 1t.\leq 3r.A_2\}$$

Applying NR3 to Clause 7 to resolve the final nested expression with r ($Z_4 \in N_C$ is a fresh definer):

$$\{4.\ \leq 1r.Z_2 \sqcup Z_1 \quad 5.\ Z_2 \sqcup \leq 1r^{-}.A_1 \quad 6.\ \neg Z_1 \sqcup \geq 2s.Z_3 \sqcup \leq 0t.A_3$$
$$8.\ \neg Z_3 \sqcup \leq 1t.Z_4 \quad 9.\ Z_4 \sqcup \neg(\leq 3r.A_2)\}$$

After simplifying the negated literals in Clauses 5 and 9, all clauses containing r or r^{-} are now in their canonical form. The resulting $\mathcal{O}'$, which is in r-NF, consists of the final set of generated clauses.

Lemma 1. *Let $\mathcal{O}$ be an $\mathcal{ALCQI}(\nabla)$-ontology and $r \in N_R$. Then $\mathcal{O}$ can be transformed into an r-NF ontology $\mathcal{O}'$ by a linear number of applications of the normalization rules NR1–NR3. The size of the resulting ontology $\mathcal{O}'$ is linear in the size of $\mathcal{O}$.*

Lemma 2. *Let $\mathcal{O}$ be an $\mathcal{ALCQI}(\nabla)$-ontology, and $\mathcal{O}'$ the normalized one obtained from $\mathcal{O}$ by applying the rules NR1–NR3. Let $sig_D(\mathcal{O}')$ denote the set of definers introduced in $\mathcal{O}'$. Then, for any interpretation $\mathcal{I}'$, the following holds:*

$$\mathcal{I}' \models \mathcal{O}' \text{ iff } \mathcal{I} \models \mathcal{O},$$

for some interpretation $\mathcal{I}$ such that $\mathcal{I} \sim_{sig_D(\mathcal{O}')} \mathcal{I}'$.

Definers are auxiliary symbols that fall outside the desired signature and must be eliminated from the final result, a process described in Section 5.

4 Single Name Elimination

Once an ontology is in r-NF, our method eliminates the role name r (along with its inverse r^-) by applying a unified inference rule, IR, detailed in Figure 1. This rule follows a premise-conclusion schema: it replaces all clauses containing r or r^- with a new set of clauses free of these roles. The fundamental idea of IR is to generalize the binary resolution principle [25, 26]. In the $\mathcal{ALCQI}$ context, this requires a unified view of polarity. An occurrence of the role name r is considered *positive* if its first-order translation contains a positive literal for the predicate $r(\cdot,\cdot)$, and *negative* otherwise. This leads to two unified sets of premises:

- **The Unified Positive Set $\mathcal{P}_{\mathbf{pos}}$**: Clauses with positive polarity for r. This set is the union of $\mathcal{P}^+(r)$ (clauses of the form $C \sqcup {\geq}mr.D$) and $\mathcal{P}^-(r^-)$ (clauses of the form $E' \sqcup {\leq}n'r^-.F'$).
- **The Unified Negative Set $\mathcal{P}_{\mathbf{neg}}$**: Clauses with negative polarity for r. This set is the union of $\mathcal{P}^-(r)$ (clauses of the form $E \sqcup {\leq}nr.F$) and $\mathcal{P}^+(r^-)$ (clauses of the form $C' \sqcup {\geq}m'r^-.D'$).

IR resolves combinations of clauses from $\mathcal{P}_{\text{pos}}$ against combinations from $\mathcal{P}_{\text{neg}}$. This unified approach is crucial as it captures all four types of interactions: forward-vs-forward (${\geq}r$. vs. ${\leq}r$.), inverse-vs-inverse (${\geq}r^-$. vs. ${\leq}r^-$.), and the two types of cross-interactions between forward and inverse roles (${\geq}r$. vs. ${\geq}r^-$., and ${\leq}r^-$. vs. ${\leq}r$.).

Let $|\mathcal{P}_{\text{pos}}| = m$ and $|\mathcal{P}_{\text{neg}}| = n$. A complete resolution requires considering all $2^m \times 2^n$ combinations between the two sets. This many-to-many relationship is simplified into two one-to-many relationships, as detailed in the comprehensive rule in Figure 1. The rule is organized into *tiers*, where the k-th tier represents the resolution involving a combination of k clauses. The novelty of our calculus lies in defining the specific resolvents for all combinations, including the challenging cross-interaction cases.

Example 2. *Consider an $\mathcal{ALCQI}$-ontology $\mathcal{O}'$ with $\mathcal{F} = \{r\}$:*

$$\mathcal{O}' = \{\ 1.\ A_1 \sqcup {\geq}1r.B_1\ (\in \mathcal{P}_{pos}),\quad 2.\ A_2 \sqcup {\leq}0r.B_2\ (\in \mathcal{P}_{neg}),$$
$$3.\ A_3 \sqcup {\geq}1r^-.B_3\ (\in \mathcal{P}_{neg}),\quad 4.\ A_4 \sqcup {\leq}0r^-.B_4\ (\in \mathcal{P}_{pos})\}$$

Applying the IR involves resolving clauses from $\mathcal{P}_{pos} = \{1, 4\}$ against clauses from $\mathcal{P}_{neg} = \{2, 3\}$. This includes four types of interactions:
1. Intra-Role Interactions (Familiar Cases):

- ***Forward vs. Forward (${\geq}r$. vs. ${\leq}r$.):*** *Resolve Clause 1 with Clause 2. This is a standard cardinality clash. Since $1 > 0$, we obtain:*

$$5.\ A_1 \sqcup A_2 \sqcup {\geq}1\nabla.(B_1 \sqcap \neg B_2).$$

- ***Inverse vs. Inverse (${\leq}r^-$. vs. ${\geq}r^-$.):*** *Resolve Clause 4 (positive polarity) with Clause 3 (negative polarity). This is symmetric to the forward case and yields:*

$$6.\ A_4 \sqcup A_3 \sqcup {\geq}1\nabla.(B_3 \sqcap \neg B_4).$$

2. Cross-Role Interactions (Novel Contribution):

- ***Forward vs. Inverse (${\geq}r$. vs. ${\geq}r^-$.):*** *Resolve Clause 1 ($\in \mathcal{P}_{pos}$) with Clause 3 ($\in \mathcal{P}_{neg}$). This is the crucial cross-interaction. Clause 1 states that any instance of A_1 has an r-successor in B_1. Clause 3 states that any instance of A_3 has an r-predecessor in B_3. If an instance of B_1 is also an instance of A_3, then a path is formed. The calculus captures this constraint propagation. A simplified resolvent generated by our specialized rule would look like:*

$$7.\ A_1 \sqcup A_3 \sqcup \neg{\geq}1\nabla.(B_1 \sqcap B_3),$$

  which states that if something is both A_1 and A_3, then B_1 and B_3 must be disjoint. This is a new constraint, free of r, derived from the interaction.
- ***Inverse vs. Forward (${\leq}r^-$. vs. ${\leq}r$.):*** *Resolve Clause 4 ($\in \mathcal{P}_{pos}$) with Clause 2 ($\in \mathcal{P}_{neg}$). This is the symmetric cross-interaction, which yields a similar type of new constraint.*

The final forgetting result $\mathcal{V}$ is the union of all such resolvents (including 0th-tier consequences not shown here), demonstrating the necessity of the unified approach to ensure completeness.

Lemma 3. *Let $\mathcal{O}'$ be an $\mathcal{ALCQI}(\nabla)$-ontology in r-NF, serving as the premises of the unified IR rule, and $\mathcal{V}$ be its conclusion. Then, for any interpretation $\mathcal{I}'$, the following holds:*

$$\mathcal{I}' \models \mathcal{O}' \text{ iff } \mathcal{I} \models \mathcal{V},$$

for some interpretation $\mathcal{I}$ such that $\mathcal{I} \sim_r \mathcal{I}'$.

Proof Sketch. The proof consists of two directions, showing that the set of premises $\mathcal{P}_r \subseteq \mathcal{O}'$ (clauses containing r or r^-) and the set of resolvents $\mathcal{R} \subseteq \mathcal{V}$ are r-equivalent.

1. Soundness (⇒ direction): We show that if $\mathcal{I}' \models \mathcal{P}_r$, then $\mathcal{I}' \models \mathcal{R}$. The proof proceeds by contradiction for each resolvent. Consider a canonical resolvent from Component A of the IR rule, generated from premises $C_i \sqcup {\geq}x_i r.D_i$ and $\{E_{j_l} \sqcup {\leq}y_{j_l} r.F_{j_l}\}_{l=1\ldots k}$. The resolvent is:

$$Res = C_i \sqcup \bigsqcup_{l=1}^{k} E_{j_l} \sqcup {\geq}(x_i - \sum_{l=1}^{k} y_{j_l})\nabla.(D_i \sqcap \bigsqcap_{l=1}^{k} \neg F_{j_l}).$$

Assume the premises are true in $\mathcal{I}'$ but Res is false. This implies the existence of an element $a \in \Delta^{\mathcal{I}'}$ falsifying each disjunct in Res. From the premises, it follows that $a \in ({\geq}x_i r.D_i)^{\mathcal{I}'}$ and $a \in ({\leq}y_{j_l} r.F_{j_l})^{\mathcal{I}'}$ for all l. By cardinality counting, a must have at least $x_i - \sum_{l=1}^{k} y_{j_l}$ successors via r in the set $(D_i \sqcap \bigsqcap_{l=1}^{k} \neg F_{j_l})^{\mathcal{I}'}$. This implies a satisfies the final disjunct of the resolvent, yielding a clash. The argument extends to all tiers and cross-role interactions, proving soundness.

2. Completeness (⇐ direction): We show that for any model $\mathcal{I} \models \mathcal{R}$, we can construct an interpretation $\mathcal{I}' \sim_r \mathcal{I}$ such that $\mathcal{I}' \models \mathcal{P}_r$. The construction defines the interpretation of $r^{\mathcal{I}'}$ while preserving all other interpretations from $\mathcal{I}$. For each $a \in \Delta^{\mathcal{I}}$, the set of resolvents $\mathcal{R}$ provides sufficient constraints to guarantee the existence of enough individuals to act as r-successors and r-predecessors. A detailed counting argument, based on the lower bounds on set cardinalities provided by the resolvents, shows that

The Unified Inference Rule IR for Eliminating Role Name r

Let $\mathcal{P}_{\text{pos}} = \{C_i \sqcup L_i^+\}_{i=1\ldots m}$ and $\mathcal{P}_{\text{neg}} = \{E_j \sqcup L_j^-\}_{j=1\ldots n}$, where L^+ and L^- are literals with positive and negative polarity for r, respectively. The conclusion is the union of all resolvents generated by the two main components below.

Component A: Resolving one Positive Clause with Combinations of Negative Clauses

For each positive clause $C_i \sqcup L_i^+$ (where L_i^+ is $\geq x_i r.D_i$ or $\leq x_i r^-.D_i$), the set of resolvents $\textsc{Resolve}(\mathcal{P}_{\text{neg}}, C_i \sqcup L_i^+)$ is the union of the following tiers:

- **0th-tier:** The r-free consequence of the clause itself. For $L_i^+ = \geq x_i r.D_i$, this yields $\{C_i \sqcup \geq x_i \nabla.D_i\}$. A symmetric rule applies for $L_i^+ = \leq x_i r^-.D_i$, yielding $\{C_i \sqcup \leq x_i \nabla.D_i\}$.
- **k-tier** ($1 \leq k \leq n$)**:** For each combination of k negative clauses $\{E_{j_1} \sqcup L_{j_1}^-, \ldots, E_{j_k} \sqcup L_{j_k}^-\} \subseteq \mathcal{P}_{\text{neg}}$, a resolvent is generated. The exact formula depends on the specific forms of L_i^+ and the $L_{j_l}^-$ literals. For the canonical interaction where $L_i^+ = \geq x_i r.D_i$ and $L_{j_l}^- = \leq y_{j_l} r.F_{j_l}$, the resolvent is:
$$\left\{C_i \sqcup \bigsqcup_{l=1}^{k} E_{j_l} \sqcup \geq (x_i - \sum_{l=1}^{k} y_{j_l}) \nabla.(D_i \sqcap \bigsqcap_{l=1}^{k} \neg F_{j_l})\right\}, \quad \text{if } x_i > \sum_{l=1}^{k} y_{j_l}.$$
The rules for cross-interactions (e.g., $\geq r.$ vs. $\geq r^-.$) constitute a core technical contribution of this work.

Component B: Resolving one Negative Clause with Combinations of Positive Clauses

For each negative clause $E_j \sqcup L_j^-$ (where L_j^- is $\leq y_j r.F_j$ or $\geq y_j r^-.F_j$), the set of resolvents $\textsc{Resolve}(\mathcal{P}_{\text{pos}}, E_j \sqcup L_j^-)$ is the union of the following tiers for $k \in \{2, \ldots, m\}$:

- For each combination of k positive clauses $\{C_{i_1} \sqcup L_{i_1}^+, \ldots, C_{i_k} \sqcup L_{i_k}^+\} \subseteq \mathcal{P}_{\text{pos}}$:
  - If k is **even**, the resolvent has the form:
$$E_j \sqcup \bigsqcup_{l=1}^{k} C_{i_l} \sqcup \bigcup_{1 \leq l_1 < l_2 \leq k} \geq z_{i_{l_1}, i_{l_2}} \nabla.(D_{i_{l_1}} \sqcap D_{i_{l_2}}) \sqcup \cdots \sqcup \geq z_{i_1,\ldots,i_k} \nabla.(\bigsqcap_{l=1}^{k} D_{i_l}) \sqcup \geq (\cdots)\nabla.(\cdots)$$
  - If k is **odd**, the resolvent has the form:
$$E_j \sqcup \bigsqcup_{l=1}^{k} C_{i_l} \sqcup \bigcup_{1 \leq l_1 < l_2 \leq k} \geq z_{i_{l_1}, i_{l_2}} \nabla.(D_{i_{l_1}} \sqcap D_{i_{l_2}}) \sqcup \cdots \sqcup \leq z_{i_1,\ldots,i_k} \nabla.(\bigsqcap_{l=1}^{k} D_{i_l}) \sqcup \geq (\cdots)\nabla.(\cdots)$$

  Here, the concepts D_{i_l} and the final cardinality restriction $\geq(\cdots)\nabla.(\cdots)$ are derived from the specific forms of the literals $L_{i_l}^+$ and L_j^-. The integer variables z are bound within specific ranges to ensure soundness. The full, explicit formulae are extensive but follow a systematic construction based on generalized cardinality constraints.

Figure 1: The unified inference rule IR. It operates on unified polarity sets $\mathcal{P}_{\text{pos}}$ and $\mathcal{P}_{\text{neg}}$ to handle all four types of resolution interactions: forward-vs-forward, inverse-vs-inverse, and the two cross-interactions between forward and inverse roles.

one can always populate $r^{\mathcal{I}'}$ to satisfy all $\geq$-constraints from $\mathcal{P}_{\text{pos}}$ without violating any $\leq$-constraints from $\mathcal{P}_{\text{neg}}$. The higher-tier resolvents are crucial for ensuring this consistency when multiple positive constraints apply simultaneously. This constructive proof guarantees the existence of a model $\mathcal{I}'$ for the premises. □

Lemma 4. *Let $m = |\mathcal{P}_{pos}|$ and $n = |\mathcal{P}_{neg}|$. The number of derived clauses by the unified IR rule has a double exponential upper bound in the worst case, related to m and n.*

Proof Sketch. The proof is based on counting the maximum number of non-isomorphic resolvents generated by the IR rule. The complexity is dominated by Component B of the rule (resolving one negative clause against combinations of positive clauses). Let us analyze the number of clauses generated from one negative clause Neg_j resolved against combinations of m positive clauses from $\mathcal{P}_{\text{pos}}$. The rule generates resolvents for each k-tier, where $k \in \{2, \ldots, m\}$.

(1) For a given tier k, there are $\binom{m}{k}$ ways to choose k positive clauses.
(2) For each such combination, the resolvent contains new literals with integer variables (the z variables), e.g., $\geq z_{i_1,i_2} \nabla.(D_{i_1} \sqcap D_{i_2})$. The number of distinct z variables in a k-tier resolvent is $2^k - 1$ (the number of non-empty subsets of the k chosen clauses).
(3) Let $z_{\max}$ be the maximum cardinality in any positive clause. The number of distinct values for each z variable is bounded by $z_{\max}$. Therefore, the number of non-isomorphic resolvents for a single combination of k positive clauses is bounded by $(z_{\max})^{2^k}$. The double-exponential nature arises here: the number of these new integer variables grows exponentially with k (as 2^k), and this term itself appears in an exponent.
(4) The total number of clauses for a single negative clause is thus bounded by:
$$\sum_{k=2}^{m} \binom{m}{k} \cdot (z_{\max})^{2^k}.$$
The dominant term occurs when $k = m$, which gives a complexity related to $(z_{\max})^{2^m}$.

Algorithm 1 The Unified Forgetting Process for $\mathcal{ALCQI}(\nabla)$

Require: An $\mathcal{ALCQI}(\nabla)$-ontology $\mathcal{O}$, a forgetting signature $\mathcal{F}$
Ensure: An $\mathcal{ALCQI}(\nabla)$-ontology $\mathcal{V}$ such that $\mathrm{sig}(\mathcal{V}) \cap \mathcal{F} = \emptyset$.

function Forget($\mathcal{O}, \mathcal{F}$)
  $\mathcal{O}_c \leftarrow$ Clausify($\mathcal{O}$) ▷ Convert all GCIs into clauses
  $\mathcal{V} \leftarrow$ Simplify($\mathcal{O}_c$) ▷ Apply simplifications
  $\mathcal{D}_{total} \leftarrow \emptyset$ ▷ Initialize the set of all introduced definers.
  ▷ **Phase 1: Iterative Role Elimination**
  for each role name $r \in \mathcal{F}$ **do**
    $(\mathcal{O}_n, \mathcal{D}_r) \leftarrow$ Normalize($\mathcal{V}, r$) ▷ Transform into r-NF, collect new definers $\mathcal{D}_r$.
    $\mathcal{O}_f \leftarrow$ EliminateRole($\mathcal{O}_n, r$) ▷ Apply IR
    $\mathcal{D}_{total} \leftarrow \mathcal{D}_{total} \cup \mathcal{D}_r$ ▷ Aggregate definers
    $\mathcal{V} \leftarrow$ Simplify($\mathcal{O}_f$) ▷ Simplify intermediate result.
  end for
  ▷ **Phase 2: Definer Elimination**
  if $\mathcal{D}_{total} \neq \emptyset$ **then**
    for each definer $Z \in \mathcal{D}_{total}$ **do**
      $\mathcal{V} \leftarrow$ Ackermann($\mathcal{V}, Z$) ▷ Eliminate definers
      $\mathcal{V} \leftarrow$ Simplify($\mathcal{V}$)
    end for
  end if
  return $\mathcal{V}$
end function

(5) Since there are n negative clauses, the total number of resolvents from Component B is bounded by $O(n \cdot 2^m \cdot (z_{\max})^{2^m})$.

Component A contributes a term that is only single exponential ($O(m \cdot 2^n)$). The total complexity is therefore dominated by Component B, which is double exponential in m, the size of the positive set. This finite upper bound proves termination. □

5 The Forgetting Process

The entire forgetting procedure is orchestrated by the process outlined in Algorithm 1. The algorithm takes an $\mathcal{ALCQI}(\nabla)$-ontology and a set of role names to forget as input and proceeds in two main phases. In Phase 1, it iteratively eliminates each specified role name by first transforming the ontology into a normal form and then applying the unified inference rule IR. In Phase 2, it eliminates the auxiliary definer concepts introduced during normalization. Throughout this process, equivalence-preserving simplifications are eagerly applied to maintain conciseness. A key feature is the flexibility of the elimination sequence, allowing role names to be forgotten in any user-specified order.

5.1 Ackermann-Based Definer Elimination

The final phase of the process is to eliminate the auxiliary definers (e.g., $Z_1, Z_2, \ldots$) introduced during normalization. This is achieved using a generalization of *Ackermann's Lemma* [40], which essentially allows for the safe substitution of a definer with its definition.

The intuition is as follows: if a definer Z is introduced to stand for a complex concept C, its definition will appear as a set of clauses like $\{\neg Z \sqcup C_1, \ldots, \neg Z \sqcup C_n\}$. If all other occurrences of Z in the ontology are positive, we can simply replace every occurrence of Z with the concept it defines, which is $C_1 \sqcap \ldots \sqcap C_n$. This effectively resolves the definer away. A dual rule applies if the definitions are positive and the occurrences are negative.

Example 3. *Suppose after the role elimination phase, we have an intermediate ontology $\mathcal{V}_{int}$ containing the definer Z:*

$$\mathcal{V}_{int} = \{1.\ \neg Z \sqcup (A \sqcap {\leq}0s.\neg B),\ 2.\ C \sqsubseteq {\geq}3t.Z,\ 3.\ G \sqsubseteq Z,\ 4.\ E \sqsubseteq F\}$$

Here, Clause 1 provides the definition for Z, corresponding to the axiom $Z \sqsubseteq (A \sqcap {\leq}0s.\neg B)$. The rest of the ontology where Z appears (Clauses 2 and 3) is positive with respect to Z. The conditions for Ackermann's Lemma are met, allowing for elimination by substitution. We replace all occurrences of Z in Clauses 2 and 3 with its full definition from Clause 1. This step ensures that the definer's meaning is propagated to all parts of the ontology where it was used. This substitution is guaranteed to be model-preserving, satisfying the conditions for strong forgetting with respect to the definer Z. After substituting and removing the original definition clause, the final, Z-free ontology is:

$$\mathcal{V}_{final} = \{2'.\ C \sqsubseteq {\geq}3t.(A \sqcap {\leq}0s.\neg B),\ 3'.\ G \sqsubseteq (A \sqcap {\leq}0s.\neg B),\ 4.\ E \sqsubseteq F\}$$

The formal statement of this principle is given by the following generalized lemma; this result follows from [40].

Lemma 5 (Soundness of Ackermann's Lemma). *Let $\mathcal{O}$ be an ontology containing the clauses $C_1 \sqcup \neg A, \ldots, C_n \sqcup \neg A$, where $A \in N_C$, and the C_i are concepts free of A. If the remaining part of the ontology, $\mathcal{O} \setminus \{C_1 \sqcup \neg A, \ldots, C_n \sqcup \neg A\}$, is positive w.r.t. A, then the ontology $\mathcal{O}^A_{C_1 \sqcap \ldots \sqcap C_n}$, obtained by replacing every occurrence of A in $\mathcal{O}$ with $C_1 \sqcap \ldots \sqcap C_n$, is a result of strongly forgetting A from $\mathcal{O}$. A dual version applies for positive definition clauses and negative occurrences.*

5.2 Limitations and Termination

While the Ackermann-based approach is typically successful, there is no guarantee of completely removing all definers, particularly in cases involving cyclic dependencies [13]. Consider forgetting r from $\mathcal{O} = \{{\geq}1r.{\leq}0r.\bot\}$, which exhibits an implicit cycle. Normalizing this yields $\mathcal{O}' = \{{\geq}1r.Z, \neg Z \sqcup {\leq}0r.\bot\}$. Applying IR produces $\{{\geq}1\nabla.Z, \neg Z \sqcup {\geq}1\nabla.Z\}$, which has an explicit cyclic dependency on Z (i.e., $Z \sqsubseteq {\geq}1\nabla.Z$). Attempting to eliminate Z via substitution would lead to an infinite regress, producing an infinite set of axioms $\mathcal{V} = \{\overbrace{{\geq}1\nabla.\cdots{\geq}1\nabla}^{n}.\top \mid n \geq 1\}$, which cannot be finitely represented in standard DLs.

Other systems like LETHE and FAME exploit fixpoint operators [4] to finitely represent such cyclic results. However, since fixpoints are not part of the standard OWL 2 DL profile and are not supported by common reasoners or the OWL API,[1] our method prioritizes practical compatibility. It terminates the process even if some definers involved in ineliminable cycles remain in the final result.

Theorem 1. *Given any $\mathcal{ALCQI}(\nabla)$-ontology $\mathcal{O}$ and any forgetting signature $\mathcal{F}$, our forgetting method (Algorithm 1) always terminates and returns an $\mathcal{ALCQI}(\nabla)$-ontology $\mathcal{V}$.*

(i) *If $\mathcal{V}$ does not contain any definers, it is a result of strongly forgetting $\mathcal{F}$ from $\mathcal{O}$.*

(ii) *If $\mathcal{V}$ contains definers (due to ineliminable cycles), it is a result of strongly forgetting $\mathcal{F} \cup \mathrm{sig}_D(\mathcal{V})$ from $\mathcal{O}$.*

[1] The universal role ∇ is supported by the OWL API.

Theorem 1 states the termination and soundness of our method. Full completeness is not achievable due to the inherent undecidability of the strong forgetting problem [12]. Nevertheless, as our empirical evaluation shows, these theoretical limitations rarely impede practical performance on real-world ontologies. In practice, the typical use case involves $\mathcal{ALCQI}$-ontologies as input, since real-world ontologies conforming to OWL 2 DL do not employ the universal role ∇ explicitly. The generality of the method ensures that it can also be applied iteratively: if a prior forgetting step produces an $\mathcal{ALCQI}(\nabla)$-ontology, subsequent forgetting operations remain well-defined.

6 Empirical Evaluation

We implemented a prototype of our approach in Java using OWL API Version 5.1.7.[2] To validate its practical applicability, we conducted extensive comparative experiments against two state-of-the-art forgetting systems—LETHE[3] and FAME[4]—using two large-scale real-world ontology corpora widely used as benchmarks in the knowledge management community. The Oxford ISG Repository[5] includes diverse OWL ontologies across multiple knowledge domains. The March 2017 snapshot from BioPortal[6] [23] comprises specialized biomedical ontologies that are central to biomedical knowledge management and health informatics.

From Oxford ISG, we initially selected 488 ontologies containing $\leq$10,000 GCIs each. To ensure meaningful evaluation of our number restriction forgetting capabilities, we filtered out ontologies containing only trivial number restrictions. Specifically, since $\exists r.C$ can be equivalently expressed as $\geq 1r.C$ and $\forall r.C$ as $\leq 0r.\neg C$, we excluded ontologies where all number restrictions are of these trivial forms. After this rigorous filtering, 196 ontologies remained eligible for evaluation. We subsequently extracted the $\mathcal{ALCQI}$ fragments from these ontologies by removing axioms not expressible or that cannot be encoded in $\mathcal{ALCQI}$. This refinement process resulted in an average GCI reduction of 4.6%. Similarly, from BioPortal, we applied the same filtering criteria to exclude ontologies with only trivial number restrictions, yielding 142 ontologies for evaluation.

To enable granular performance analysis across ontologies of varying size, we stratified the Oxford ontologies into three size-based partitions maintaining proportional distribution, where |Onto| denotes the number of GCIs in an ontology:

- I: 142 small-scale ontologies with $10 \leq |\text{Onto}| < 1000$
- II: 43 medium-scale ontologies with $1000 \leq |\text{Onto}| < 4999$
- III: 11 large-scale ontologies with $5000 \leq |\text{Onto}| < 10000$

Applying identical filtering and stratification criteria to the 142 BioPortal ontologies, we partitioned them as:

- I: 88 small-scale ontologies with $10 \leq |\text{Onto}| < 1000$
- II: 45 medium-scale ontologies with $1000 \leq |\text{Onto}| < 4999$
- III: 9 large-scale ontologies with $5000 \leq |\text{Onto}| < 10000$

Detailed distributional statistics of all test ontologies are provided in the extended version.

[2]http://owlcs.github.io/owlapi/
[3]https://lat.inf.tu-dresden.de/ koopmann/LETHE/
[4]https://www.cs.man.ac.uk/ schmidt/sf-fame/
[5]http://krr-nas.cs.ox.ac.uk/ontologies/lib/
[6]https://bioportal.bioontology.org/

The forgetting signature $\mathcal{F}$ varies significantly with task-specific requirements. To comprehensively assess our approach, we configured three experimental settings eliminating 10%, 30%, and 50% of role names from each ontology's signature, aligning with established benchmarking practices [14, 34, 36, 42]. These percentages reflect a spectrum of realistic use cases: 10% corresponds to light-weight tasks such as ontology debugging or computing logical differences between ontology versions where few symbols change; 30% models intermediate scenarios such as ontology analysis and abduction; and 50% represents aggressive simplification for knowledge summarization or cross-domain reuse. Names in $\mathcal{F}$ were randomly selected via shuffling to ensure unbiased evaluation.

All experiments were run on a laptop with Intel Core i7-9750H (6 cores, 2.70 GHz) and 12 GB DDR4-1600 MHz RAM, under strict resource constraints: 600-second timeout and 10GB heap space. An experiment was *successful* if: (i) all role names in $\mathcal{F}$ were eliminated; (ii) no definers remained in output; (iii) completion within timeout; and (iv) memory usage below limit. Each experiment was repeated 100 times with averaged metrics reported for statistical robustness.

We note that our forgetting algorithm is fully *deterministic*: given the same ontology $\mathcal{O}$ and forgetting signature $\mathcal{F}$, it always produces the identical output in the identical time. The sole source of variance across repetitions is the random sampling of $\mathcal{F}$ via shuffling, which determines *which* role names are selected for elimination. The 100 repetitions per configuration thus sample diverse forgetting tasks across the role name space of each ontology, and the averaged metrics provide stable estimates of expected performance. Since there is no stochastic component in the algorithm itself (no random initialization, no sampling-based inference, no non-deterministic search), reporting standard deviations or conducting statistical significance tests over repeated runs of the *same* $(\mathcal{O}, \mathcal{F})$ pair would be uninformative: the result is identical every time. This deterministic property is standard for symbolic reasoning algorithms in the DL and knowledge representation literature.

Tables 1 and 4 present our prototype's performance. The SR (success rate) constitutes the primary metric, quantifying successful completion percentages. Our prototype achieved average success rates of approximately 89%, 83%, and 74% for 10%, 30%, and 50% forgetting scenarios, respectively. Success rates decline as $|\mathcal{F}|$ increases due to heightened computational complexity.

Failures manifested through timeout (TR), memory overflow (MR), and definer retention (DR), all increasing with $|\mathcal{F}|$. The DR column quantifies ontologies where auxiliary definers could not be fully eliminated. A bottleneck emerged: excessive role name occurrences within number restrictions trigger combinatorial explosion in intermediate clauses. For instance, BioPortal's SDO ontology contains the role 'hasPart' in over 50 $\geq$-restrictions. Forgetting such roles demands processing up to 2^{50} clauses, causing memory exhaustion within the 9GB constraint. A detailed analysis of how role coupling degree affects forgetting performance, together with a scalability study across ontology sizes, is provided in the extended version.

As the first method for strong role forgetting in $\mathcal{ALCQI}(\nabla)$, our approach has no direct baseline: no other implemented system supports strong forgetting with both qualified number restrictions and inverse roles. The closest available tools are LETHE [41] and FAME [43], which represent the state of the art for forgetting in less

Table 1: Our Prototype's Results on Oxford-ISG ($\mathcal{ALCQI}$)

$\mathcal{F}$%	Part	Time	Mem	SR	TR	MR	DR	DC
10%	I	2.89	700.77	91.27	0.56	2.11	6.06	1.09
	II	4.18	908.99	83.52	1.39	4.63	10.46	4.36
	III	5.82	1067.43	78.40	0.40	16.80	4.40	6.57
	Avg	**3.29**	**760.45**	**88.96**	**0.82**	**3.07**	**7.15**	**2.09**
30%	I	4.27	994.19	88.16	0.99	4.51	6.34	3.41
	II	7.47	1273.21	71.69	4.35	14.81	9.15	12.49
	III	11.65	1607.70	62.72	8.64	20.64	8.00	16.31
	Avg	**5.17**	**1071.14**	**83.19**	**1.97**	**7.57**	**7.27**	**6.08**
50%	I	9.91	1695.68	83.15	2.11	6.34	8.40	5.71
	II	12.40	2081.54	49.94	5.56	19.44	25.06	22.47
	III	17.61	2567.12	43.12	12.88	28.56	15.44	34.67
	Avg	**10.51**	**1780.75**	**73.71**	**3.31**	**10.01**	**12.97**	**10.90**

Table 2: LETHE's Results on Oxford-ISG ($\mathcal{ALC}$ fragments)

$\mathcal{F}$%	Part	Time	Mem	SR	TR	MR	DR	DC
10%	I	9.89	1010.11	87.60	3.52	3.24	5.64	35.96
	II	10.40	1343.74	79.85	3.24	7.13	9.78	48.57
	III	49.06	2005.42	58.80	4.40	32.80	4.00	66.36
	Avg	**12.00**	**1134.93**	**84.34**	**3.49**	**5.61**	**6.66**	**40.31**
30%	I	22.13	1330.09	82.82	5.21	5.92	6.05	135.41
	II	42.23	1835.33	66.24	5.09	21.30	7.37	152.67
	III	71.09	2043.54	47.04	8.48	36.48	8.00	170.36
	Avg	**29.09**	**1478.46**	**77.33**	**5.40**	**10.74**	**6.53**	**141.02**
50%	I	36.27	2415.34	75.08	8.09	8.72	8.11	259.30
	II	78.06	2962.44	48.10	8.81	23.65	19.44	273.13
	III	139.80	3581.89	31.36	12.64	44.00	12.00	270.20
	Avg	**50.83**	**2596.18**	**67.49**	**8.47**	**13.99**	**10.05**	**262.92**

Table 3: FAME's Results on Oxford-ISG ($\mathcal{ALC}$ fragments)

$\mathcal{F}$%	Part	Time	Mem	SR	TR	MR	DR	DC
10%	I	5.97	764.54	90.10	2.39	1.83	5.68	9.57
	II	9.41	999.70	80.76	4.12	5.05	10.07	13.19
	III	23.61	1653.94	58.80	4.40	28.80	8.00	32.11
	Avg	**6.11**	**894.12**	**86.58**	**2.88**	**4.01**	**6.53**	**11.53**
30%	I	10.65	934.58	85.03	3.52	5.37	6.08	25.56
	II	22.21	1259.83	68.96	4.07	19.26	7.71	39.57
	III	49.52	2039.57	47.04	12.48	32.48	8.00	43.66
	Avg	**11.60**	**1106.26**	**79.60**	**4.09**	**10.20**	**6.11**	**29.08**
50%	I	21.89	1342.76	77.60	6.62	7.57	8.21	58.22
	II	34.42	1972.94	53.54	7.87	19.15	19.44	79.46
	III	55.01	2573.02	43.12	8.32	36.56	12.00	84.32
	Avg	**23.83**	**1618.07**	**71.09**	**6.98**	**11.48**	**10.45**	**54.77**

Table 4: Our Prototype's Results on BioPortal ($\mathcal{ALCQI}$)

$\mathcal{F}$%	Part	Time	Mem	SR	TR	MR	DR	DC
10%	I	1.70	564.86	96.53	0.25	0.99	2.23	0.34
	II	3.02	787.66	86.79	1.44	4.33	7.44	1.83
	III	4.90	938.95	49.00	10.50	25.50	15.00	24.50
	Avg	**2.21**	**645.52**	**90.39**	**1.32**	**3.67**	**4.62**	**1.60**
30%	I	3.04	859.78	91.58	0.50	2.48	5.44	1.39
	II	5.85	1097.97	83.17	1.44	5.29	10.10	4.05
	III	10.33	1503.13	49.00	10.50	25.50	15.00	28.00
	Avg	**4.16**	**955.43**	**86.03**	**1.32**	**4.89**	**7.76**	**3.14**
50%	I	6.29	1169.74	78.71	1.49	4.46	15.34	0.75
	II	8.85	1819.12	75.48	2.40	7.12	15.00	6.15
	III	14.82	2262.83	39.20	10.80	25.00	25.00	37.50
	Avg	**7.38**	**1412.53**	**75.21**	**2.44**	**6.83**	**15.52**	**3.65**

Table 5: LETHE's Results on BioPortal ($\mathcal{ALC}$ fragments)

$\mathcal{F}$%	Part	Time	Mem	SR	TR	MR	DR	DC
10%	I	7.95	725.85	90.84	1.98	4.95	2.23	21.21
	II	11.43	954.22	83.89	3.37	5.30	7.44	42.96
	III	32.39	1169.43	49.00	10.50	25.50	15.00	61.76
	Avg	**10.56**	**825.91**	**86.04**	**2.94**	**6.32**	**4.62**	**30.64**
30%	I	18.81	1128.76	85.93	4.46	4.17	5.44	65.19
	II	35.32	1529.12	78.37	5.29	6.24	10.10	116.32
	III	57.49	1973.81	49.00	10.50	25.50	15.00	193.23
	Avg	**26.45**	**1308.32**	**81.17**	**5.12**	**6.11**	**7.76**	**89.25**
50%	I	32.82	1696.08	74.75	5.45	7.91	11.89	166.21
	II	67.26	2081.54	69.71	8.17	9.12	13.00	205.85
	III	131.95	2567.12	39.20	10.80	25.00	25.00	286.12
	Avg	**49.88**	**1780.75**	**70.99**	**6.73**	**9.48**	**12.80**	**186.21**

Table 6: FAME's Results on BioPortal ($\mathcal{ALC}$ fragments)

$\mathcal{F}$%	Part	Time	Mem	SR	TR	MR	DR	DC
10%	I	2.36	647.16	91.33	2.97	3.47	2.23	5.64
	II	8.64	871.06	85.87	3.37	3.32	7.44	12.07
	III	19.95	993.57	49.00	10.50	25.50	15.00	16.20
	Avg	**5.44**	**733.10**	**87.01**	**2.94**	**5.41**	**4.64**	**8.34**
30%	I	7.69	1013.13	87.38	3.96	3.22	5.44	16.41
	II	15.59	1227.51	80.29	5.29	4.32	10.10	27.05
	III	28.40	1625.05	49.00	10.50	25.50	15.00	48.65
	Avg	**11.48**	**1118.66**	**82.86**	**4.89**	**4.89**	**7.36**	**21.78**
50%	I	13.24	1294.76	77.72	5.94	4.45	11.89	42.50
	II	19.86	1456.13	71.64	9.12	6.24	13.00	50.40
	III	32.30	1664.68	39.20	10.80	25.00	25.00	64.25
	Avg	**18.96**	**1368.94**	**73.46**	**7.23**	**6.33**	**12.98**	**46.36**

expressive logics. LETHE implements weak forgetting for $\mathcal{ALCI}$, while FAME implements strong forgetting for $\mathcal{ALCOIH}$; neither supports qualified number restrictions ($\mathcal{Q}$). Our comparative evaluation is therefore inherently asymmetric: we process the strictly more expressive $\mathcal{ALCQI}$ ontologies while LETHE and FAME operate on the $\mathcal{ALC}$ fragments of the same corpora. This asymmetry works *against* our method, making any performance advantage we achieve all the more significant. To further address this asymmetry, Section 6.5 presents an additional comparison where our prototype is also evaluated on the same $\mathcal{ALC}$ fragments.

Tables 2, 3, 5, and 6 present comparative results. Despite processing more expressive $\mathcal{ALCQI}$ ontologies versus competitors' $\mathcal{ALCI}$ fragments, our prototype maintained consistent 2–5% success rate advantages. This gap intensifies in challenging scenarios—at 50% forgetting on Oxford Part III, our method achieved 43.12% versus LETHE's 31.36% and FAME's 43.12% while processing substantially more complex constructs.

Critically, our approach demonstrates substantial computational efficiency. Examining Time reveals our prototype executes 3.6–4.8× faster than LETHE and 1.9–2.3× faster than FAME on average. For 50% forgetting on Oxford, our method required merely 10.51 seconds versus LETHE's 50.83 and FAME's 23.83 seconds. Memory consumption (Mem) shows 33–46% reduction versus LETHE and 13–18% versus FAME, demonstrating superior space efficiency.

The DC (definer count) column reveals a fundamental algorithmic advantage: our method introduces dramatically fewer definers—an order of magnitude reduction versus both baselines. LETHE employs exponential definer introduction correlated with role restriction counts, while our approach achieves linear complexity. For instance, at 50% forgetting on Oxford, our method introduced 10.90

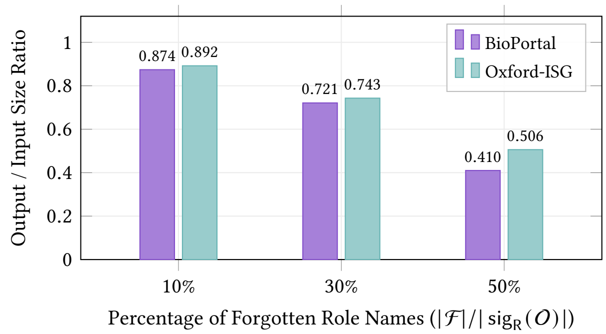

Figure 2: Output-to-input size ratio across forgetting scenarios. All ratios remain well below 1, indicating no size blowup in practice.

definers versus LETHE's 262.92 and FAME's 54.77. This reduction directly translates to faster execution and smaller memory footprints through reduced intermediate clause processing and simplified definer elimination. A comprehensive complexity analysis demonstrating how linear definer introduction contributes to these performance advantages is available in our GitHub repository.

All three methods exhibited marginally superior BioPortal performance versus Oxford, manifesting as modest improvements in success rate, execution time, and memory utilization. This differential derives from BioPortal's intrinsic structural simplicity—reduced axiom counts, lower role name density, and shallower structures versus Oxford's more heterogeneous and complex landscape.

6.1 Output Size Analysis

A natural concern with forgetting is whether the output ontology suffers from size blowup, given the triple exponential worst-case bound [22, 24]. Figure 2 reports the ratio of the number of axioms in the forgetting result to the number of axioms in the input ontology, averaged over all successful runs in each configuration.

Contrary to the worst-case theoretical prediction, no size blowup is observed in practice. All ratios remain well below 1, indicating that the forgetting result is consistently *smaller* than the input ontology. The reduction scales with the forgetting percentage: at 10% forgetting, the output retains approximately 87–89% of the input size; at 30%, approximately 72–74%; and at 50%, approximately 41–51%. This near-linear relationship reflects the fact that the number of newly derived axioms is typically small compared to the number of axioms removed during elimination. The slightly higher ratios on Oxford-ISG are consistent with its denser and more interconnected axiom structure, which tends to generate more derived consequences per eliminated role name.

6.2 Phase-Level Time Breakdown

Our forgetting pipeline consists of three sequential phases: (1) *Normalization*, which transforms the ontology into r-NF; (2) *Role Elimination (IR)*, which applies the inference rule to eliminate each target role name; and (3) *Definer Elimination*, which removes auxiliary definers introduced during normalization. Figure 3 reports the average percentage of total execution time consumed by each phase.

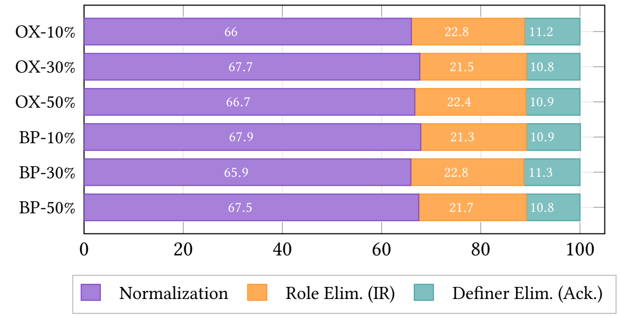

Figure 3: Phase-level time breakdown. OX = Oxford-ISG, BP = BioPortal. Normalization consistently dominates execution time across all configurations.

Table 7: Impact of inverse roles on forgetting performance. Ontologies containing inverse role declarations (w/ $\mathcal{I}$) exhibit moderately lower success rates and higher execution times.

Dataset	Group	#	SR_{10}	SR_{30}	SR_{50}	$Time_{10}$	$Time_{50}$
Oxford	w/ $\mathcal{I}$	95	86.2	79.8	70.1	3.89	12.73
	w/o $\mathcal{I}$	101	91.5	86.3	77.1	2.73	8.92
BioPortal	w/ $\mathcal{I}$	42	87.6	82.4	71.2	2.71	9.15
	w/o $\mathcal{I}$	100	91.6	87.5	77.9	2.00	6.64

Across all configurations, the time distribution is remarkably stable: normalization consumes approximately 66–68% of the total execution time, role elimination via IR accounts for 21–23%, and Ackermann-based definer elimination takes roughly 11%. This consistency holds across both corpora and all three forgetting scenarios. The dominance of normalization is expected, as transforming each complex clause into r-NF requires systematic decomposition of nested expressions, introducing fresh definers for each sub-concept containing the target role. The relatively modest share of IR reflects the efficiency of our inference calculus, which resolves all relevant interactions in a single pass per role name. The small Ackermann overhead confirms that the definers introduced during normalization are, in most cases, straightforwardly eliminable through the standard Ackermann procedure.

6.3 Impact of Inverse Roles

A distinguishing feature of our method is its ability to handle inverse roles ($\mathcal{I}$) within qualified number restrictions. To quantify the practical impact of this feature, we partition the test ontologies into two groups: those whose signature contains at least one inverse role declaration (**w/** $\mathcal{I}$) and those without (**w/o** $\mathcal{I}$). Table 7 reports the success rate and average execution time for each group.

The results reveal a consistent and interpretable pattern across both corpora. Ontologies containing inverse roles exhibit 4–7% lower success rates and 24–43% longer execution times compared to those without. This overhead is attributable to the cross-interaction resolution mechanism in our inference calculus: when a role r and its inverse r^- both appear within qualified number restrictions, the IR rule must resolve four types of interactions (forward-vs-forward, inverse-vs-inverse, and two cross-orientations), compared to only

Table 8: Frequency of universal role ∇ in forgetting results. The vast majority of outputs remain within $\mathcal{ALCQI}$.

$\mathcal{F}$%	Metric	Oxford-ISG		BioPortal	
		w/ ∇	Avg #	w/ ∇	Avg #
10%		8.3%	1.4	4.6%	0.9
30%		14.7%	3.2	9.2%	2.1
50%		21.5%	5.8	15.8%	4.3

two types in the absence of inverse roles. The additional resolvents increase intermediate clause counts, raising the probability of timeout or memory overflow.

Notably, the overhead remains moderate rather than dramatic. This is because the IR phase constitutes only about 22% of total execution time (Section 6.2), and not every role in the forgetting signature $\mathcal{F}$ necessarily has an inverse declaration. The proportion of ontologies containing inverse roles is substantially higher in Oxford-ISG (48%) than in BioPortal (30%), reflecting the greater structural diversity of general-purpose ontologies compared to domain-specific biomedical knowledge bases.

6.4 Universal Role Usage

Our method may introduce the universal role ∇ to preserve semantic completeness when the forgetting result cannot be expressed within $\mathcal{ALCQI}$ alone. Understanding how frequently this occurs is important for assessing the practical usability of the output, since results that remain within $\mathcal{ALCQI}$ can be used directly in standard OWL 2 toolchains without adaptation. Table 8 reports the percentage of successful forgetting runs whose output contains at least one axiom involving ∇, together with the average number of ∇-axioms per affected ontology.

The results confirm that the universal role is needed only in a minority of cases. At 10% forgetting, fewer than 9% of outputs on Oxford-ISG and fewer than 5% on BioPortal require ∇. Even at 50% forgetting, the majority of results (over 78% on Oxford-ISG, over 84% on BioPortal) remain entirely within $\mathcal{ALCQI}$. When ∇ does appear, it is used sparingly: affected ontologies contain on average fewer than 6 ∇-axioms even in the most aggressive forgetting scenario. The higher ∇ frequency on Oxford-ISG is consistent with its greater proportion of inverse role declarations (Section 6.3), since ∇ arises specifically from cross-interactions between a role r and its inverse r^- during elimination. These findings are practically encouraging: in the vast majority of use cases, the forgetting output can be deployed directly as a standard $\mathcal{ALCQI}$ ontology without requiring tool support for the extended logic.

6.5 Comparison Fairness

The comparative results in Tables 1–6 show that our prototype, while processing the more expressive $\mathcal{ALCQI}$ ontologies, consistently outperforms Lethe and Fame on $\mathcal{ALC}$ fragments. A natural question is whether this advantage stems from our algorithm or from processing different logic fragments. To address this, we additionally ran our prototype on the same $\mathcal{ALC}$ fragments used by Lethe and Fame. Table 9 presents the results at 50% forgetting, the most computationally demanding scenario.

Results confirm our advantages hold independently of the logic fragment. On the same $\mathcal{ALC}$ inputs, our prototype achieves 6–11% higher success rates, runs 2.9–6.2× faster than Lethe and 2.3–3.3× faster than Fame, and introduces an order of magnitude fewer definers. Comparing our own tool across the two logic settings (Ours$_{\mathcal{ALC}}$ vs. Ours$_{\mathcal{ALCQI}}$) reveals the cost of supporting $\mathcal{Q}$ and $\mathcal{I}$: a modest 3–4% decrease in success rate and approximately 22–28% increase in execution time. This overhead is commensurate with the substantially greater expressive power of $\mathcal{ALCQI}$ and confirms that our method scales gracefully to richer logics.

Table 9: Comparison fairness at 50% forgetting. "Ours$_{\mathcal{ALC}}$" denotes our prototype restricted to $\mathcal{ALC}$ fragments; "Ours$_{\mathcal{ALCQI}}$" is the full system on $\mathcal{ALCQI}$ ontologies.

Dataset	Tool	Logic	SR	Time	DC
Oxford	Lethe	$\mathcal{ALC}$	67.49	50.83	262.92
	Fame	$\mathcal{ALC}$	71.09	23.83	54.77
	Ours$_{\mathcal{ALC}}$	$\mathcal{ALC}$	77.2	8.21	7.93
	Ours$_{\mathcal{ALCQI}}$	$\mathcal{ALCQI}$	73.71	10.51	10.90
BioPortal	Lethe	$\mathcal{ALC}$	70.99	49.88	186.21
	Fame	$\mathcal{ALC}$	73.46	18.96	46.36
	Ours$_{\mathcal{ALC}}$	$\mathcal{ALC}$	78.8	5.73	2.66
	Ours$_{\mathcal{ALCQI}}$	$\mathcal{ALCQI}$	75.21	7.38	3.65

Despite addressing the more challenging $\mathcal{ALCQI}$ forgetting problem, our prototype achieves superior practical performance across all key metrics: higher success rates, significantly reduced computational time and memory requirements, and substantially more efficient definer management. The output remains compact with no size blowup (Section 6.1), normalization dominates execution time while the core inference stays efficient (Section 6.2), inverse roles impose only moderate overhead (Section 6.3), the universal role ∇ is rarely needed (Section 6.4), and our algorithmic advantages hold even under same-fragment comparison (Section 6.5). These results validate our theoretical contributions and confirm practical viability for real-world ontology forgetting applications.

7 Conclusion and Future Work

In this paper, we have developed the first method for strong forgetting of role names (also concept names via a standard reduction) in a DL with qualified number restrictions and inverse roles. This ability to compute strong forgetting results means that our approach can be applied to any syntactic fragments of $\mathcal{ALCQI}(\nabla)$, ensuring consistent results. This versatility is particularly beneficial in problems such as modal correspondence theory [28] and second-order quantifier elimination [8, 27], while addressing contemporary challenges in managing large-scale knowledge bases and supporting knowledge-intensive information systems. Since strong forgetting yields stronger results, this approach's application in ontology-based tasks, particularly in abduction, offers results richer in information than those obtained using weak forgetting approaches like [6, 15]. This makes it more powerful in identifying true causes or explanations, a crucial feature for building more transparent and trustworthy knowledge-driven applications.

Moving forward, our immediate goal is to extend our method to ABoxes, enabling the abstraction of large-scale instance data in knowledge graphs. Such a capability would directly support critical use cases like creating privacy-preserving views for knowledge sharing and data integration pipelines. Additionally, we will adapt our approach to accommodate decidable extensions of $\mathcal{ALCQI}$ [2].

GenAI Usage Disclosure

GenAI were used to assist with editing and proofreading the manuscript text. All technical content, proofs, algorithms, experimental design, and data analysis were produced entirely by the authors.

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A Missing Proofs

To prove Lemma 1, we first define some key notions. Consider X as a syntactic object including concepts, roles, and clauses. Given a role name $r \in \mathsf{N_R}$, the *frequency* $\mathsf{fq}(r, X)$ of r in X is defined inductively as follows:

- $\mathsf{fq}(r, s) = \begin{cases} 1, & \text{if } s = r, \\ 0, & \text{if otherwise;} \end{cases}$
- $\mathsf{fq}(r, A) = 0$, for any $A \in \mathsf{N_C} \cup \{\top\}$;
- $\mathsf{fq}(r, \neg D) = \mathsf{fq}(r, D)$,
- $\mathsf{fq}(r, \geq ms.D) = \mathsf{fq}(r, s) + \mathsf{fq}(r, D)$;
- $\mathsf{fq}(r, \leq ns.D) = \mathsf{fq}(r, s) + \mathsf{fq}(r, D)$;
- $\mathsf{fq}(r, E * F) = \mathsf{fq}(r, E) + \mathsf{fq}(r, F)$, for $* \in \{\sqcap, \sqcup\}$,

and its *role depth* in X, denoted as $\mathsf{rd}(r, X)$, as follows:

- $\mathsf{rd}(r, s) = 0$, for any $s \in \mathsf{N_R} \cup \{\nabla\}$;
- $\mathsf{rd}(r, A) = 0$, for any $A \in \mathsf{N_C} \cup \{\top\}$;
- $\mathsf{rd}(r, \neg D) = \mathsf{rd}(r, D)$,
- $\mathsf{rd}(r, \geq ms.D) = 1 + \mathsf{rd}(r, D)$;
- $\mathsf{rd}(r, \leq ns.D) = 1 + \mathsf{rd}(r, D)$;
- $\mathsf{rd}(r, E * F) = max(\mathsf{rd}(r, E), \mathsf{rd}(r, F))$, for $* \in \{\sqcap, \sqcup\}$.

Proposition 1. *A clause X containing r or r^- is in r-NF iff $\mathsf{fq}(r, X) = 1$ and $\mathsf{rd}(r, X) = 1$.*

Proof. ($\Rightarrow$) Suppose X is in r-NF. By Definition 2, X has the form $C \sqcup \geq mR.D$ or $E \sqcup \leq nR.F$ with $R \in \{r, r^-\}$, where C, D, E, F contain no occurrence of r or r^-. Hence r occurs exactly once in X, namely in the role position R, so $\mathsf{fq}(r, X) = 1$. Moreover, this single occurrence lies directly under one number restriction and the filler is r-free, so $\mathsf{rd}(r, X) = 1$.

($\Leftarrow$) Suppose $\mathsf{fq}(r, X) = 1$ and $\mathsf{rd}(r, X) = 1$. Since X contains r or r^- and $\mathsf{fq}(r, X) = 1$, there is exactly one occurrence of r. Since $\mathsf{rd}(r, X) = 1$, this occurrence lies under exactly one number restriction and not in a nested filler. Thus the occurrence must be in the role position $R \in \{r, r^-\}$ of a top-level literal $\geq mR.D$ or $\leq nR.F$, with the remaining disjuncts and the filler being r-free. This is precisely the r-NF of Definition 2. □

Lemma 1. *Let $\mathcal{O}$ be an $\mathcal{ALCQI}(\nabla)$-ontology. Then $\mathcal{O}$ can be transformed into r-NF $\mathcal{O}'$ by a linear number of applications of the normalization rules NR1–NR3. In addition, the size of the resulting ontology $\mathcal{O}'$ is linear in the size of $\mathcal{O}$.*

Proof. By Proposition 1, a clause X containing r or r^- is in r-NF iff $\mathsf{fq}(r, X) = 1$ and $\mathsf{rd}(r, X) = 1$. Each application of NR1–NR3 to a clause X not in r-NF replaces an r-containing subconcept by a fresh definer and emits one auxiliary clause; this strictly decreases $\mathsf{fq}(r, X)$ or $\mathsf{rd}(r, X)$ by at least one while keeping both non-negative. Since both measures are bounded by the size of $\mathcal{O}$, the process terminates after a linear number of applications. Each application introduces exactly one definer and one auxiliary clause, so the number of definers and new clauses is bounded by $O(n)$, where n is the number of number restrictions in $\mathcal{O}$. Hence the size of $\mathcal{O}'$ is linear in the size of $\mathcal{O}$. □

Lemma 2. *Let $\mathcal{O}$ be an $\mathcal{ALCQI}(\nabla)$-ontology, and $\mathcal{O}'$ the normalized one obtained from $\mathcal{O}$ by applying the rules NR1–NR3. Let $\mathsf{sig}_D(\mathcal{O}')$ denote the set of definers introduced in $\mathcal{O}'$. Then, for any interpretation $\mathcal{I}'$, the following holds:*

$$\mathcal{I}' \models \mathcal{O}' \text{ iff } \mathcal{I} \models \mathcal{O},$$

for some interpretation $\mathcal{I}$ such that $\mathcal{I} \sim_{\mathsf{sig}_D(\mathcal{O}')} \mathcal{I}'$.

Proof. Observe that the application of any of the normalization rules results in the introduction of a single fresh definer $Z \in \mathsf{N_C}$. We show that for any new ontology $\mathcal{O}'$ obtained from the ontology $\mathcal{O}$ by applying any of the normalization rules, the following holds: for any interpretation $\mathcal{I}'$,

$$\mathcal{I}' \models \mathcal{O}' \text{ iff } \mathcal{I} \models \mathcal{O},$$

for some interpretation $\mathcal{I}$ such that $\mathcal{I} \sim_Z \mathcal{I}'$. We treat NR1 in detail. NR2 and NR3 can be proved in a similar way.

In examining NR1, let us consider that $\mathcal{O}'$ is derived from $\mathcal{O}$ by substituting the clause $C \sqcup \geq mr.D$ (with $r \in \mathsf{sig}(C)$) with two new clauses: $Z \sqcup \geq mr.D$ and $\neg Z \sqcup C$, where $Z \in \mathsf{N_C}$ is a fresh definer. Evidently, $\mathsf{sig}(\mathcal{O}')$ extends $\mathsf{sig}(\mathcal{O})$ by including Z.

Consider $\mathcal{I}'$ as a model of $\mathcal{O}'$, then $Z^{\mathcal{I}'} \cup (\geq mr.D)^{\mathcal{I}'}$ and $(\neg Z)^{\mathcal{I}'} \cup C^{\mathcal{I}'}$ hold. By resolving these two clauses on Z, we obtain that $C^{\mathcal{I}'} \cup (\geq mr.D)^{\mathcal{I}'}$ holds. Since $\mathcal{I} \sim_Z \mathcal{I}'$ and $Z \notin \mathsf{sig}(C \sqcup \geq mr.D)$, we have $C^{\mathcal{I}} \cup (\geq mr.D)^{\mathcal{I}}$, confirming that $\mathcal{I}$ is a model of $\mathcal{O}$.

Conversely, if $\mathcal{I}$ is a model of $\mathcal{O}$, let $\mathcal{I}'$ be an interpretation identical to $\mathcal{I}$ on all names except for Z, where $Z^{\mathcal{I}'}$ is defined as $C^{\mathcal{I}}$. Since $\mathcal{I}$ is a model of $\mathcal{O}$, $C^{\mathcal{I}} \cup (\geq mr.D)^{\mathcal{I}}$ holds. Since $Z \notin \mathsf{sig}(C \sqcup \geq mr.D)$, we have $C^{\mathcal{I}} = C^{\mathcal{I}'}$ and $(\geq mr.D)^{\mathcal{I}} = (\geq mr.D)^{\mathcal{I}'}$. By construction $Z^{\mathcal{I}'} = C^{\mathcal{I}'}$, which makes both $Z^{\mathcal{I}'} \cup (\geq mr.D)^{\mathcal{I}'}$ and $(\neg Z)^{\mathcal{I}'} \cup C^{\mathcal{I}'}$ true. Therefore, $\mathcal{I}'$ is a model of $\mathcal{O}'$.

The cases for NR2 and NR3 are analogous. For NR3, which replaces F in a clause $E \sqcup \leq nr.F$ (with $r \in \mathsf{sig}(F)$) and adds $Z \sqcup \neg F$, the resolution step combines $E \sqcup \leq nr.Z$ and $Z \sqcup \neg F$ on Z; setting $Z^{\mathcal{I}'} = F^{\mathcal{I}}$ in the converse direction establishes the equivalence in the same manner. □

Lemma 3. *Let $\mathcal{O}'$ be an $\mathcal{ALCQI}(\nabla)$-ontology in r-NF, serving as the premises of the IR rule, and $\mathcal{V}$ be its conclusion. Then, for any interpretation $\mathcal{I}'$, the following holds:*

$$\mathcal{I}' \models \mathcal{O}' \text{ iff } \mathcal{I} \models \mathcal{V},$$

for some interpretation $\mathcal{I}$ such that $\mathcal{I} \sim_r \mathcal{I}'$.

Proof. We denote all r-clauses in the premises of this rule as $\mathcal{P}$ and the conclusion, excluding $\mathcal{O}^{-r}$, as $\mathcal{C}$. Lemma 3 states r-equivalence of $\mathcal{O}'$ and $\mathcal{V}$. Given that $\mathcal{O}^{-r}$ is not involved in the inference process, proving r-equivalence of $\mathcal{P}$ and $\mathcal{C}$ would suffice, that is, for any interpretation $\mathcal{I}'$,

$$\mathcal{I}' \models \mathcal{P} \text{ iff } \mathcal{I} \models \mathcal{C},$$

for some interpretation $\mathcal{I}$ such that $\mathcal{I} \sim_r \mathcal{I}'$.

The "only if" direction: Given that $\mathcal{I} \sim_r \mathcal{I}'$, it suffices to show that if any model $\mathcal{I}' \models \mathcal{P}$, then $\mathcal{I}' \models \mathcal{C}$. This ensures that $\mathcal{I} \models \mathcal{C}$ as well, as $r \notin \mathsf{sig}(\mathcal{C})$.

Recall from the unified polarity view of Section 4 that a clause $E' \sqcup \leq n'r^-.F'$ has the same (positive) polarity for r as a clause $C \sqcup \geq mr.D$, and a clause $C' \sqcup \geq m'r^-.D'$ has the same (negative) polarity as $E \sqcup \leq nr.F$. Consequently, the four interaction types (forward-vs-forward, inverse-vs-inverse, and the two forward-vs-inverse cross-interactions) are resolved by a single arithmetic argument on the number of r-edges incident to a domain element, where inverse literals constrain incoming r-edges and forward literals constrain outgoing r-edges. We present the argument for the canonical forward-vs-forward case; the remaining cases follow by the same counting, reading r^--restrictions as constraints on the in-degree rather than the out-degree.

We first examine the set $\textsc{Resolve}(\mathcal{P}^-(r), C_i \sqcup \geq x_i r.D_i)$. Clearly, $C_i \sqcup \geq x_i r.D_i \models C_i \sqcup \geq x_i \nabla.D_i$ (0th-tier), then we have $\mathcal{I}' \models C_i \sqcup \geq x_i \nabla.D_i$.

Consider the case $C_i \sqcup E_{j_1} \sqcup \cdots \sqcup E_{j_k} \sqcup \geq (x_i - y_{j_1} - \cdots - y_{j_k})\nabla.(D_i \sqcap \neg F_{j_1} \sqcap \cdots \sqcap \neg F_{j_k})$ (1st-tier to nth-tier). Suppose there exists a domain element $a \in \Delta^{\mathcal{I}'}$ such that $a \notin C_i^{\mathcal{I}'}, a \notin E_{j_1}^{\mathcal{I}'}, \cdots, a \notin E_{j_k}^{\mathcal{I}'}$, then we have $a \in (\geq x_i r.D_i)^{\mathcal{I}'}, a \in (\leq y_{j_1} r.F_{j_1})^{\mathcal{I}'}, \cdots, a \in (\leq y_{j_k} r.F_{j_k})^{\mathcal{I}'}$. If the number of r-successors of a in $(D_i \sqcap \neg F_{j_1} \sqcap \cdots \sqcap \neg F_{j_k})^{\mathcal{I}'}$ is less than $x_i - y_{j_1} - \cdots - y_{j_k}$, then there must be at least $y_{j_1} + \cdots + y_{j_k} + 1$ r-successors of a in $(F_{j_1} \sqcup \cdots \sqcup F_{j_k})^{\mathcal{I}'}$, which contradicts $a \in (\leq y_{j_1} r.F_{j_1})^{\mathcal{I}'}, \cdots, a \in (\leq y_{j_k} r.F_{j_k})^{\mathcal{I}'}$. Therefore we have $\mathcal{I}' \models C_i \sqcup E_{j_1} \sqcup \cdots \sqcup E_{j_k} \sqcup \geq (x_i - y_{j_1} - \cdots - y_{j_k})\nabla.(D_i \sqcap \neg F_{j_1} \sqcap \cdots \sqcap \neg F_{j_k})$.

We then examine the set $\textsc{Resolve}(\mathcal{P}^+(r), E_j \sqcup \leq y_j r.F_j)$. For convenience, we refer to this set as C'

For any element $a \in \Delta^{\mathcal{I}'}$, let us examine its relation to $E_j^{\mathcal{I}'}$:

- If $a \in E_j^{\mathcal{I}'}$, then for any clause $\alpha \in C'$, we have $a \in \alpha^{\mathcal{I}'}$;
- If $a \notin E_j^{\mathcal{I}'}$, then suppose that $a \notin C_1^{\mathcal{I}'}, \cdots, a \notin C_k^{\mathcal{I}'}$ and $a \in C_{k+1}^{\mathcal{I}'}, \cdots, a \in C_m^{\mathcal{I}'}$. For any clause $\alpha \in C'$, if α contains C_t, for $t > k$, then $a \in \alpha^{\mathcal{I}'}$. Thus, our focus narrows to those clauses in C' that exclude C_t. In a more generalized context, consider a clause β containing $C_1, \cdots, C_k$. For a literal $l_\geq$ of the form $\geq Z\nabla.D$, if $a \notin l_\geq^{\mathcal{I}'}$, it indicates $Z > |D^{\mathcal{I}'}|$; similarly, for a literal $l_\leq$ of the form $\leq Z\nabla.D$, if $a \notin l_\leq^{\mathcal{I}'}$, it indicates $Z < |D^{\mathcal{I}'}|$. Considering the extreme case where, for all literals $l_\geq$ and $l_\leq$ in β, $a \notin l_\geq^{\mathcal{I}'}$ and $a \notin l_\leq^{\mathcal{I}'}$, the requirement for $a \in \beta^{\mathcal{I}'}$ to hold requires that:

$$a \in \Big(\geq \big(\sum_{i=1}^{k} x_i - \sum_{1 \leq j_1 < j_2 \leq k} (Z_{j_1,j_2} - 1) + \sum_{1 \leq j_1 < j_2 < j_3 \leq k} (Z_{j_1,j_2,j_3} + 1) + \cdots + (-1)^{k+1} Z_{1,\cdots,k} + 1 - y_j \big)\nabla.((D_{i_1} \sqcup \cdots \sqcup D_{i_k}) \sqcap \neg F_j) \Big)^{\mathcal{I}'}$$

When examining the above literal, which involves the $\geq$-restrictions, it suffices to demonstrate that a satisfies the belonging condition when the cardinality of $\geq$ takes its maximum value. This is because $Z_{j1,j2} > |(D_{j1} \sqcap D_{j2})^{\mathcal{I}'}|$ and $Z_{j1,j2,j3} < |(D_{j1} \sqcap D_{j2} \sqcap D_{j3})^{\mathcal{I}'}|$. Hence, the upper limit S_k for this cardinality is:

$$S_k = \sum_{i=1}^{k} x_i - \sum_{1 \leq j1 < j2 \leq k} |(D_{j1} \sqcap D_{j2})^{\mathcal{I}'}| + \sum_{1 \leq j_1 < j_2 < j_3 \leq k} |(D_{j1} \sqcap D_{j2} \sqcap D_{j3})^{\mathcal{I}'}| + \cdots + (-1)^{k+1} |(D_1 \sqcap \cdots \sqcap D_k)^{\mathcal{I}'}| - y_j$$

Building on our initial assumption, where a is a domain element in $(\geq x_i r.D_i)^{\mathcal{I}'}$, it follows that a has at least x_i r-successors in each $D_i^{\mathcal{I}'}$ for $1 \leq i \leq k$. To understand how S_k is derived, let us consider the r-successors in the combined set $D_1^{\mathcal{I}'} \cup D_2^{\mathcal{I}'}$. While $D_1^{\mathcal{I}'}$ and $D_2^{\mathcal{I}'}$ each have at x_1 and x_2 r-successors, respectively, the union $(D_1 \sqcup D_2)^{\mathcal{I}'}$ may not have exactly $x_1 + x_2$ r-successors due to overlaps. In fact, by the inclusion-exclusion principle, $(D_1 \sqcup D_2)^{\mathcal{I}'}$ contains at least $x_1 + x_2 - |(D_1 \sqcap D_2)^{\mathcal{I}'}|$ r-successors. Extending this idea, $(D_1 \sqcup D_2 \sqcup D_3)^{\mathcal{I}'}$ has at least $x_1 + x_2 + x_3 - |(D_1 \sqcap D_2)^{\mathcal{I}'}| - |(D_1 \sqcap D_3)^{\mathcal{I}'}| - |(D_2 \sqcap D_3)^{\mathcal{I}'}| + |(D_1 \sqcap D_2 \sqcap D_3)^{\mathcal{I}'}|$ r-successors. Thus, $S_k + y_j$ denotes the minimum count of r-successors in $(D_1 \sqcup \cdots \sqcup D_k)^{\mathcal{I}'}$.

To prove that $a \in (\geq S_k \nabla.((D_1 \sqcup \cdots \sqcup D_k) \sqcap \neg F_j))^{\mathcal{I}'}$, we use a proof by contradiction. Suppose $|((D_1 \sqcup \cdots \sqcup D_k) \sqcap \neg F_j)^{\mathcal{I}'}| < S_k$. This implies that $((D_1 \sqcup \cdots \sqcup D_k) \sqcap F_j)^{\mathcal{I}'}$ contains at least $y_j + 1$ r-successors, contradicting $a \in (\leq y_j r.F_j)^{\mathcal{I}'}$. Hence, $|((D_1 \sqcup \cdots \sqcup D_k) \sqcap \neg F_j)^{\mathcal{I}'}| \geq S_k$, and $a \in (\geq S_k \nabla.((D_1 \sqcup \cdots \sqcup D_k) \sqcap \neg F_j))^{\mathcal{I}'}$.

The "if" direction: For any model $\mathcal{I}$ of C, we can always extend this interpretation w.r.t. the role name r to construct a new interpretation $\mathcal{I}'$ such that $\mathcal{I}' \models \mathcal{P}$. $\mathcal{I}$ and $\mathcal{I}'$ have the same domain, i.e., $\Delta^{\mathcal{I}} = \Delta^{\mathcal{I}'}$, and differ only possibly in how they interpret r, while their interpretations of all other names remain identical. For an element $a \in \Delta^{\mathcal{I}'}$, the construction of $r^{\mathcal{I}'}$ depends on whether a is an element of $E_j^{\mathcal{I}'}$:

- If $a \notin E_j^{\mathcal{I}'}$ and suppose that $a \notin C_1^{\mathcal{I}'}, \cdots, a \notin C_k^{\mathcal{I}'}$ and $a \in C_{k+1}^{\mathcal{I}'}, \cdots, a \in C_m^{\mathcal{I}'}$, it is clear that $a \in (C_{k+1} \sqcup \geq x_{k+1} r.D_{k+1})^{\mathcal{I}'}, \cdots, a \in (C_m \sqcup \geq x_m r.D_m)^{\mathcal{I}'}$. Given that $\mathcal{I} \models C$, this implies $a \in (\geq x_1 \nabla.D_1)^{\mathcal{I}'}, \cdots, a \in (\geq x_k \nabla.D_k)^{\mathcal{I}'}$, which further implies that for any $1 \leq i \leq k$, $|D_i^{\mathcal{I}'}| \geq x_i$. Our goal is to designate x_i elements from each $D_i^{\mathcal{I}'}$ $(1 \leq i \leq k)$ as r-successors. This ensures that no more than y_i among these belong to $F_j^{\mathcal{I}'}$. Since $|D_i^{\mathcal{I}'}| \geq x_i$, the first part of this goal is fulfilled. The following discusses how to achieve the second part which concerns the restriction on elements belonging to $F_j^{\mathcal{I}'}$.

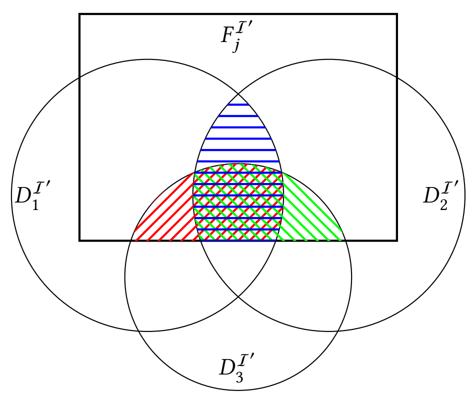

Figure 4: Relations among $D_1^{\mathcal{I}'}$, $D_2^{\mathcal{I}'}$, $D_3^{\mathcal{I}'}$, and $F_j^{\mathcal{I}'}$

To provide clarity and enhance comprehension, we exemplify our proof strategy with the specific case where $k = 3$. In this case, the relationships among $D_1^{\mathcal{I}'}$, $D_2^{\mathcal{I}'}$, $D_3^{\mathcal{I}'}$, and $F_j^{\mathcal{I}'}$ can be visually conceptualized by a Venn diagram, as depicted in Figure 4. Here, the rectangular area at the top represents $F_j^{\mathcal{I}'}$, while three circles positioned on the left, right, and bottom represent $D_1^{\mathcal{I}'}$, $D_2^{\mathcal{I}'}$, and $D_3^{\mathcal{I}'}$, respectively. Recall the meaning of S_k; specifically, when $k = 3$, S_3 is defined as:

$$S_3 = x_1 + x_2 + x_3 - |D_1^{\mathcal{I}'} \cap D_2^{\mathcal{I}'}| - |D_1^{\mathcal{I}'} \cap D_3^{\mathcal{I}'}| - |D_2^{\mathcal{I}'} \cap D_3^{\mathcal{I}'}| + |D_1^{\mathcal{I}'} \cap D_2^{\mathcal{I}'} \cap D_3^{\mathcal{I}'}| - y_j$$

Building on the analysis from the preceding section of the proof, we deduce that $a \in (\geq S_3 \nabla.((D_1 \sqcup D_2 \sqcup D_3) \sqcap \neg F))^{\mathcal{I}'}$, indicating that $|((D_1 \sqcup D_2 \sqcup D_3) \sqcap \neg F)^{\mathcal{I}'}| \geq S_3$. Applying the inclusion-exclusion principle, we can derive the following conclusion:

$$\begin{aligned}
&|((D_1 \sqcup D_2 \sqcup D_3) \sqcap \neg F)^{\mathcal{I}'}| \\
=&|D_1^{\mathcal{I}'} \cap \neg F_j^{\mathcal{I}'}| + |D_2^{\mathcal{I}'} \cap \neg F_j^{\mathcal{I}'}| + |D_3^{\mathcal{I}'} \cap \neg F_j^{\mathcal{I}'}| \\
&- |D_1^{\mathcal{I}'} \cap D_2^{\mathcal{I}'} \cap \neg F_j^{\mathcal{I}'}| - |D_1^{\mathcal{I}'} \cap D_3^{\mathcal{I}'} \cap \neg F_j^{\mathcal{I}'}| \\
&- |D_2^{\mathcal{I}'} \cap D_3^{\mathcal{I}'} \cap \neg F_j^{\mathcal{I}'}| \\
&+ |D_1^{\mathcal{I}'} \cap D_2^{\mathcal{I}'} \cap D_3^{\mathcal{I}'} \cap \neg F_j^{\mathcal{I}'}| \geq S_3
\end{aligned} \tag{1}$$

In Figure 4, the region marked with horizontal lines represents the intersection $D_1^{\mathcal{I}'} \cap D_2^{\mathcal{I}'} \cap F_j^{\mathcal{I}'}$, while the regions with left and right diagonal lines represent $D_1^{\mathcal{I}'} \cap D_3^{\mathcal{I}'} \cap F_j^{\mathcal{I}'}$ and $D_2^{\mathcal{I}'} \cap D_3^{\mathcal{I}'} \cap F_j^{\mathcal{I}'}$, respectively. Finally, the region where all three line styles converge represents $D_1^{\mathcal{I}'} \cap D_2^{\mathcal{I}'} \cap D_3^{\mathcal{I}'} \cap \neg F_j^{\mathcal{I}'}$.

In selecting elements from $D_i^{\mathcal{I}'}$, our first step is to choose all elements from $(D_i \sqcap \neg F_j)^{\mathcal{I}'}$. Consequently, we need to select only $x_i - |(D_i \sqcap \neg F_j)^{\mathcal{I}'}|$ elements from $(D_i \sqcap F_j)^{\mathcal{I}'}$. The total count of elements to be chosen from $F_j^{\mathcal{I}'}$, not accounting for other conditions, is $x_1 - |(D_1 \sqcap \neg F_j)^{\mathcal{I}'}| + x_2 - |(D_2 \sqcap \neg F_j)^{\mathcal{I}'}| + x_3 - |(D_3 \sqcap \neg F_j)^{\mathcal{I}'}|$, which we denote as M. As depicted in Figure 4, overlaps may exist between $(D_1 \sqcap F_j)^{\mathcal{I}'}$, $(D_2 \sqcap F_j)^{\mathcal{I}'}$, and $(D_3 \sqcap F_j)^{\mathcal{I}'}$. Therefore, in selecting elements from $(D_1 \sqcap F_j)^{\mathcal{I}'}$, prioritizing the overlapping regions can reduce the total number of needed elements. Elements in these overlapping regions need to be selected only once, necessitating a deduction of the duplicated portions from M. By the inclusion-exclusion principle, the adjusted count of elements chosen from $F_j^{\mathcal{I}'}$ is $M - |(D_1 \sqcap D_2 \sqcap F)^{\mathcal{I}'}| - |(D_1 \sqcap D_3 \sqcap F)^{\mathcal{I}'}| - |(D_2 \sqcap D_3 \sqcap F)^{\mathcal{I}'}| + |(D_1 \sqcap D_2 \sqcap D_3 \sqcap F)^{\mathcal{I}'}|$. Rewriting Equation 1, we obtain the following formula:

$$\begin{aligned}
&x_1 - |(D_1 \sqcap \neg F_j)^{\mathcal{I}'}| + x_2 - |(D_2 \sqcap \neg F_j)^{\mathcal{I}'}| + x_3 \\
&- |(D_3 \sqcap \neg F_j)^{\mathcal{I}'}| - |D_1^{\mathcal{I}'} \cap D_2^{\mathcal{I}'}| - |D_1^{\mathcal{I}'} \cap D_3^{\mathcal{I}'}| \\
&- |D_2^{\mathcal{I}'} \cap D_3^{\mathcal{I}'}| + |D_1^{\mathcal{I}'} \cap D_2^{\mathcal{I}'} \cap D_3^{\mathcal{I}'}| + |D_1^{\mathcal{I}'} \cap D_2^{\mathcal{I}'} \cap \neg F_j^{\mathcal{I}'}| \\
&+ |D_1^{\mathcal{I}'} \cap D_3^{\mathcal{I}'} \cap \neg F_j^{\mathcal{I}'}| + |D_2^{\mathcal{I}'} \cap D_3^{\mathcal{I}'} \cap \neg F_j^{\mathcal{I}'}| \\
&- |D_1^{\mathcal{I}'} \cap D_2^{\mathcal{I}'} \cap D_3^{\mathcal{I}'} \cap \neg F_j^{\mathcal{I}'}| \leq y_j
\end{aligned} \tag{2}$$

Upon examining Equation 1, we notice that the intersection $D_1^{\mathcal{I}'} \cap D_2^{\mathcal{I}'}$ comprise two distinct segments. The first segment falls within $F_j^{\mathcal{I}'}$, represented as $D_1^{\mathcal{I}'} \cap D_2^{\mathcal{I}'} \cap F_j^{\mathcal{I}'}$, while the second falls outside $F_j^{\mathcal{I}'}$, represented as $D_1^{\mathcal{I}'} \cap D_2^{\mathcal{I}'} \cap \neg F_j^{\mathcal{I}'}$. Consequently, Equation 2 can be further expressed as follows:

$$\begin{aligned}
&x_1 - |(D_1 \sqcap \neg F_j)^{\mathcal{I}'}| + x_2 - |(D_2 \sqcap \neg F_j)^{\mathcal{I}'}| \\
&+ x_3 - |(D_3 \sqcap \neg F_j)^{\mathcal{I}'}| - |(D_1 \sqcap D_2 \sqcap F)^{\mathcal{I}'}| \\
&- |(D_1 \sqcap D_3 \sqcap F)^{\mathcal{I}'}| - |(D_2 \sqcap D_3 \sqcap F)^{\mathcal{I}'}| \\
&+ |(D_1 \sqcap D_2 \sqcap D_3 \sqcap F)^{\mathcal{I}'}| \leq y_j
\end{aligned} \tag{3}$$

In Equation 3, the left-hand side quantifies the number of elements required to be chosen from $F_j^{\mathcal{I}'}$. The analysis, illustrated with $k = 3$ as an example, establishes that selecting no more than y_j elements from $F_j^{\mathcal{I}'}$ suffices to fulfill the condition $a \in (\geq x_1 r.D_1)^{\mathcal{I}'}, \cdots, a \in (\geq x_m r.D_m)^{\mathcal{I}'}$. This finding is applicable to scenarios where k assumes any value. For any k, we deduce that $|((D_1 \sqcup \cdots \sqcup D_k) \sqcap \neg F)^{\mathcal{I}'}| \geq S_k$. Applying the inclusion-exclusion principle, we can conclude the following:

$$\begin{aligned}
&|((D_1 \sqcup \cdots \sqcup D_k) \sqcap \neg F)^{\mathcal{I}'}| \\
&= \sum_{1 \leq i \leq k} |(D_i \sqcap \neg F_j)^{\mathcal{I}'}| \\
&- \sum_{1 \leq i_1 < i_2 \leq k} |(D_{i_1} \sqcap D_{i_2} \sqcap \neg F_j)^{\mathcal{I}'}| + \\
&\sum_{1 \leq i_1 < i_2 < i_3 \leq k} |(D_{i_1} \sqcap D_{i_2} \sqcap D_{i_3} \sqcap \neg F_j)^{\mathcal{I}'}| + \cdots \\
&+ (-1)^{k+1} |(D_1 \sqcap D_2 \sqcap \cdots \sqcap D_k \sqcap \neg F_j)^{\mathcal{I}'}|
\end{aligned} \tag{4}$$

Disregarding other conditions, the total number of elements required to be selected from $F_j^{\mathcal{I}'}$ is given by:

$$M = \sum_{1 \leq i \leq k} (x_i - |(D_i \sqcap \neg F_j)^{\mathcal{I}'}|).$$

but when accounting for potential overlaps among the sets, the inclusion-exclusion principle modifies this total. Hence, the actual number of elements selected from $F_j^{\mathcal{I}'}$ is determined as follows:

$$\begin{aligned}
&M - \sum_{1 \leq i_1 < i_2 \leq k} |(D_{i_1} \sqcap D_{i_2} \sqcap F_j)^{\mathcal{I}'}| \\
&+ \sum_{1 \leq i_1 < i_2 < i_3 \leq k} |(D_{i_1} \sqcap D_{i_2} \sqcap D_{i_3} \sqcap F_j)^{\mathcal{I}'}| \\
&+ \cdots + (-1)^{k+1} |(D_1 \sqcap D_2 \sqcap \cdots \sqcap D_k \sqcap F_j)^{\mathcal{I}'}|
\end{aligned} \tag{5}$$

Considering that Eq. 4 is greater than or equal to S_k, we deduce that Eq. 5 is less than or equal to y_j. This implies that the number of elements selected from $F_j^{\mathcal{I}'}$ in constructing the interpretation of r does not exceed y_j. Therefore, the condition $a \in (\leq y_j r.F_j)^{\mathcal{I}'}$ always hold.

- If $a \in E_j^{\mathcal{I}'}$, it follows that $a \in (E_j \sqcup \leq y_j r.F_j)^{\mathcal{I}'}$. Consider any C_i, either $a \in C_i^{\mathcal{I}'}$ or $a \notin C_i^{\mathcal{I}'}$. Assuming $a \notin C_1^{\mathcal{I}'}, \cdots, a \notin C_k^{\mathcal{I}'}$ but $a \in C_{k+1}^{\mathcal{I}'}, \cdots, a \in C_m^{\mathcal{I}'}$, we can deduce that $a \in (C_{k+1} \sqcup \geq x_{k+1} r.D_{k+1})^{\mathcal{I}'}, \cdots, a \in (C_m \sqcup \geq x_m r.D_m)^{\mathcal{I}'}$. Since $a \notin C_1^{\mathcal{I}'}, \cdots, a \notin C_k^{\mathcal{I}'}$, it implies that $a \in (\geq x_1 \nabla.D_1)^{\mathcal{I}'}, \cdots, a \in (\geq x_k \nabla.D_k)^{\mathcal{I}'}$, leading to that $|D_1^{\mathcal{I}'}| \geq x_1, \cdots, |D_k^{\mathcal{I}'}| \geq x_k$. In the case where only the negative r-clause $E_j \sqcup \leq y_j r.F_j$ is considered, all elements in $D_1^{\mathcal{I}'} \cup \cdots \cup D_k^{\mathcal{I}'}$ can be taken as r-successors of a. Since $|D_1^{\mathcal{I}'}| \geq x_1, \cdots, |D_k^{\mathcal{I}'}| \geq x_k$, it follows that $a \in (\geq x_1 r.D_1)^{\mathcal{I}'}, \cdots, a \in (\geq x_k r.D_k)^{\mathcal{I}'}$, and hence $a \in (C_1 \sqcup \geq x_1 r.D_1)^{\mathcal{I}'}, \cdots, a \in (C_k \sqcup \geq x_k r.D_k)^{\mathcal{I}'}$. However, for other negative r-clauses $E_i \sqcup \leq y_i r.F_i$ $(i \neq j)$, this construction may result in the number of r-successors of a in

$F_i^{\mathcal{I}'}$ exceeding y_i. Consequently, if $a \notin E_i^{\mathcal{I}'}$, it follows that $a \notin (E_i \sqcup \leq y_i r.F_i)^{\mathcal{I}'}$.

To avoid the above situation when constructing the interpretation of r, it is necessary to fully consider all negative r-clauses. For $E_i \sqcup \leq y_i r.F_i$ $(i \neq j)$, whenever $a \in E_i^{\mathcal{I}'}$, it holds that $a \in (E_i \sqcup \leq y_i r.F_i)^{\mathcal{I}'}$ regardless of the method used to construct the interpretation of r. Therefore, we only need to consider the case where $a \notin E_i^{\mathcal{I}'}$. For convenience, assume $a \notin E_{i_1}^{\mathcal{I}'}, \cdots, a \notin E_{i_t}^{\mathcal{I}'}$. For positive r-clauses $C_i \sqcup \geq x_i r.D_i$ $(1 \leq i \leq k)$, since $a \in (C_i \sqcup E_{i_1} \sqcup \cdots \sqcup E_{i_t} \sqcup \geq (x_i - y_{i_1} - \cdots - y_{i_t})\nabla.(D_i \sqcap \neg F_{i_1} \sqcap \cdots \sqcap \neg F_{i_t}))^{\mathcal{I}'}$, and $a \notin C_i^{\mathcal{I}'}$, it follows that $|(D_i \sqcap \neg F_{i_1} \sqcap \cdots \sqcap \neg F_{i_t})^{\mathcal{I}'}| \geq x_i - y_{i_1} - \cdots - y_{i_t}$. When constructing the interpretation of r, first select $x_i - y_{i_1} - \cdots - y_{i_t}$ elements from $(D_i \sqcap \neg F_{i_1} \sqcap \cdots \sqcap \neg F_{i_t})^{\mathcal{I}'}$ as the successors of a w.r.t. r. Then, as discussed earlier for the case $a \notin E^{\mathcal{I}'}$, for $i_1 \leq s \leq i_t$, select y_s elements from the intersection of F_s and D_i as the successors of a w.r.t. r.

This construction assigns to a enough r-successors to satisfy every positive clause $C_i \sqcup \geq x_i r.D_i$ with $a \notin C_i^{\mathcal{I}'}$, while keeping, for each negative clause $E_i \sqcup \leq y_i r.F_i$ with $a \notin E_i^{\mathcal{I}'}$, the number of r-successors in $F_i^{\mathcal{I}'}$ at most y_i. Performing this assignment independently for every $a \in \Delta^{\mathcal{I}'}$ (and symmetrically for the r^--restrictions, which constrain the in-degree) yields an interpretation $\mathcal{I}'$ with $\mathcal{I}' \sim_r \mathcal{I}$ that satisfies all clauses in $\mathcal{P}$. Hence $\mathcal{I}' \models \mathcal{P}$, which completes the "if" direction. □

Lemma 4. *Let $m = |\mathcal{P}_{pos}|$ and $n = |\mathcal{P}_{neg}|$. The number of derived clauses by the unified IR rule has a double exponential upper bound in the worst case, related to m and n.*

Proof. We count the maximum number of non-isomorphic resolvents generated by the IR rule. The conclusion is the union of the resolvent sets produced by Component A and Component B, so we bound each and take the dominant term. Let $z_{\max}$ denote the largest cardinality bound x_i or y_j occurring in any premise.

Component A. For each positive clause $C_i \sqcup L_i^+$ $(1 \leq i \leq m)$, the rule resolves it against every subset of the n negative clauses, generating one resolvent per subset. There are 2^n subsets, and the 0th-tier consequence is included. Hence Component A contributes at most $m \cdot 2^n$ resolvents. The cardinality of the ∇-restriction in each such resolvent is determined by $x_i - \sum_l y_{j_l}$, which introduces no additional multiplicity beyond the choice of subset. Component A is therefore single exponential, bounded by $O(m \cdot 2^n)$.

Component B. For each negative clause $E_j \sqcup L_j^-$ $(1 \leq j \leq n)$, the rule resolves it against combinations of k positive clauses, $k \in \{2, \ldots, m\}$. We bound the number of resolvents generated for a fixed j.

(1) For a given tier k, there are $\binom{m}{k}$ ways to choose k positive clauses.
(2) For each chosen combination, the resolvent contains one ∇-literal of the form $\geq z_T \nabla.(\bigsqcap_{t \in T} D_t)$ or $\leq z_T \nabla.(\bigsqcap_{t \in T} D_t)$ for each non-empty subset T of the k chosen clauses. The number of such subsets, and hence of distinct integer variables z_T, is $2^k - 1$.
(3) Each variable z_T ranges over the integers determined by the cardinality bounds of the involved premises, so it takes at most $z_{\max}$ distinct values. As the resolvent is determined by the joint assignment to all $2^k - 1$ variables, the number of non-isomorphic resolvents for one combination is bounded by $(z_{\max})^{2^k - 1} \leq (z_{\max})^{2^k}$. This is the source of the double exponential: the count of integer variables grows as 2^k, and this quantity appears in the exponent.
(4) Summing over all tiers, the number of resolvents generated from a single negative clause is bounded by
$$\sum_{k=2}^{m} \binom{m}{k} (z_{\max})^{2^k}.$$
The term $k = m$ dominates, contributing $(z_{\max})^{2^m}$, and $\sum_{k=2}^{m} \binom{m}{k} \leq 2^m$, so the sum is bounded by $2^m (z_{\max})^{2^m}$.
(5) Since there are n negative clauses, Component B is bounded by $O\Big(n \cdot 2^m \cdot (z_{\max})^{2^m}\Big)$.

Total. Component A contributes $O(m \cdot 2^n)$ and Component B contributes $O(n \cdot 2^m \cdot (z_{\max})^{2^m})$. The latter dominates, giving a double-exponential upper bound in m. As this bound is finite, the inference rule generates finitely many resolvents and therefore terminates. □

Theorem 1. *Given any $\mathcal{ALCQI}(\nabla)$-ontology $\mathcal{O}$ and any forgetting signature $\mathcal{F}$ of role names, our forgetting method always terminates and returns an $\mathcal{ALCQI}(\nabla)$-ontology $\mathcal{V}$.*

(i) *If $\mathcal{V}$ does not contain any definers, it is a result of forgetting $\mathcal{F}$ from $\mathcal{O}$, i.e., for any interpretation $\mathcal{I}'$, $\mathcal{I}' \models \mathcal{V}$ iff there is an interpretation $\mathcal{I} \sim_{\mathcal{F}} \mathcal{I}'$ such that $\mathcal{I} \models \mathcal{O}$.*

(ii) *If $\mathcal{V}$ contains definers, for any interpretation $\mathcal{I}'$, $\mathcal{I}' \models \mathcal{V}$ iff there is an interpretation $\mathcal{I} \sim_{\mathcal{F} \cup \mathrm{sig}_{\mathrm{D}}(\mathcal{V})} \mathcal{I}'$ such that $\mathcal{I} \models \mathcal{O}$.*

Proof. *Termination.* Algorithm 1 processes each $r \in \mathcal{F}$ once. For a fixed r, normalization terminates after linearly many steps by Lemma 1, and the inference rule IR produces at most the finitely many resolvents bounded in Lemma 4. Phase 2 eliminates each definer in $\mathcal{D}_{total}$ at most once via Ackermann's Lemma, halting on any definer whose elimination would re-enter a previously seen pattern (an ineliminable cycle). Since $\mathcal{F}$ and $\mathcal{D}_{total}$ are finite, the procedure terminates.

Correctness. Let Σ_i denote the signature after the i-th elimination step. By Lemma 2, each normalization step preserves all models modulo the introduced definers, i.e., it yields an ontology that is $\mathrm{sig}_{\mathrm{D}}$-equivalent to its input. By Lemma 3, each application of IR that eliminates a role name r yields an ontology that is r-equivalent to its input. By Lemma 5, each Ackermann step that eliminates a definer Z yields an ontology that is Z-equivalent to its input. Composing these equivalences along the run of the algorithm gives, for any interpretation $\mathcal{I}'$, that $\mathcal{I}' \models \mathcal{V}$ iff there is $\mathcal{I} \sim_{\mathcal{F} \cup \mathcal{D}_{rem}} \mathcal{I}'$ with $\mathcal{I} \models \mathcal{O}$, where $\mathcal{D}_{rem} = \mathrm{sig}_{\mathrm{D}}(\mathcal{V})$ is the set of definers that remain in $\mathcal{V}$.

(i) If $\mathcal{V}$ contains no definers, then $\mathcal{D}_{rem} = \emptyset$ and $\mathrm{sig}(\mathcal{V}) \subseteq \mathrm{sig}(\mathcal{O}) \setminus \mathcal{F}$, so the composed equivalence reduces to $\mathcal{I} \sim_{\mathcal{F}} \mathcal{I}'$. Both conditions of Definition 1 hold, and $\mathcal{V}$ is a result of forgetting $\mathcal{F}$ from $\mathcal{O}$.

(ii) If $\mathcal{V}$ contains definers, then $\mathcal{D}_{rem} \neq \emptyset$ and the composed equivalence is exactly $\mathcal{I} \sim_{\mathcal{F} \cup \mathrm{sig}_{\mathrm{D}}(\mathcal{V})} \mathcal{I}'$, so $\mathcal{V}$ is a result of forgetting $\mathcal{F} \cup \mathrm{sig}_{\mathrm{D}}(\mathcal{V})$ from $\mathcal{O}$. □

B Statistics of Test Ontologies

Table 10: Statistics of Oxford-ISG & BioPortal

Oxford ISG		**min**	**max**	**median**	**mean**	**upper decile**
	$\lvert N_C \rvert$	0	1582	86	191	545
	$\lvert N_R \rvert$	0	332	10	29	80
	$\lvert Onto \rvert$	10	990	162	262	658
	$\lvert N_C \rvert$	200	5877	1665	1769	2801
	$\lvert N_R \rvert$	0	887	11	34	61
	$\lvert Onto \rvert$	1008	4976	2282	2416	3937
	$\lvert N_C \rvert$	1162	9809	4042	5067	8758
	$\lvert N_R \rvert$	1	158	4	23	158
	$\lvert Onto \rvert$	5112	9783	7277	7195	9179
BioPortal		**min**	**max**	**median**	**mean**	**upper decile**
	$\lvert N_C \rvert$	0	784	127	192	214
	$\lvert N_R \rvert$	0	122	5	15	17
	$\lvert Onto \rvert$	10	794	283	312	346
	$\lvert N_C \rvert$	5	4530	1185	1459	1591
	$\lvert N_R \rvert$	0	131	12	30	33
	$\lvert Onto \rvert$	1023	4880	2401	2619	2782
	$\lvert N_C \rvert$	432	8340	4363	4387	4806
	$\lvert N_R \rvert$	0	135	17	30	34
	$\lvert Onto \rvert$	5457	8339	6934	6912	7109

Table 10 summarizes the distributional statistics of the test ontologies from Oxford ISG and BioPortal repositories, where $\lvert N_C \rvert$, $\lvert N_R \rvert$, and $\lvert Onto \rvert$ denote the counts of concept names, role names, and GCIs, respectively. Each corpus is stratified into three size-based partitions (I, II, III) to enable granular performance analysis across ontologies of varying complexity.